Every blanket-wafer ALD recipe hides a 40× under-dose: a chamber-to-feature model of supply-limited atomic layer deposition*

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Abstract

ALD recipes are tuned on blanket wafers. Product wafers are not blank: a 3D-NAND array of 100 nm holes at aspect ratio 100 carries 56 times the surface area of the wafer under it, and all of it has to saturate in one pulse. We built a model of a 300 mm showerhead ALD chamber in which every point on the wafer runs its own feature-scale model of a hole, coupled both ways to the chamber gas, and asked how long the pulse must be. The answer is a single rule: the saturating pulse equals 1.2 times (1 + patterned area ratio) times the time the showerhead needs to deliver one monolayer of precursor. It held within 25 % across aspect ratios 20 to 100 and a twenty-fold range of precursor feed; on non-uniform layouts the local wall-area ratio of the densest region sets the pulse, not the wafer average. At a typical operating point the blanket wafer saturates in 0.35 s and the product wafer in 14 s. Chamber pressure, showerhead gap and carrier flow do not move that number; precursor delivery rate does. Nor does the sticking coefficient across its published range; only the blanket pulse does, threefold. The water half-cycle, fifty times less reactive, needs the same pulse on the product wafer and coats it more evenly. The feature model sits within 3 to 11 % of a converged ballistic reference (Argonne's Machball and this work's own 3D Monte Carlo solver, which agree within 2 %) and fits published thickness profiles from lateral test structures, and the rule itself reproduces the measured saturation time of a fluidized-bed ALD run on 940 m^2 of silica gel to 7 %. The rule turns a day of recipe transfer into arithmetic, and says where a tool's throughput actually comes from.

What this means for a tool (one paragraph for the process engineer). If your recipe was saturated on a monitor wafer, multiply the pulse by one plus the product wafer's wall-area ratio before you trust it on product. Size the pulse on the densest or deepest region's own wall-area ratio; the wafer average under-doses it. Pressure, gap and carrier flow are uniformity knobs; they buy no time. Time comes from precursor molecules per second: vapour pressure, bubbler flow, injector design, or an architecture that exposes the wafer to a denser zone (spatial ALD wins by the same factor the rule predicts). Low-sticking reactants are not the bottleneck on patterned wafers; their cost is on blanket wafers only. And when the deepest feature saturates, the chamber shows it: the precursor stops being consumed and appears downstream. On a 3D-NAND wafer the pulses are three quarters of the cycle and the purges a quarter, the reverse of the blanket wafer, so the throughput lever is precursor delivery and the purge lever is water desorption, not chamber flow.

*Draft 0.9.3 (public), 10 November 2026. Every number is from a run documented in the repository run log, `paper/assumptions.md`.

1 Introduction

Three facts set up the problem. First, 3D NAND and DRAM are built from features with aspect ratios of 50 to 100, and ALD is the only way to coat their insides [16, 17], so the precursor must reach the bottom of every feature on the wafer inside a single pulse. Second, the patterned surface area of such a wafer is tens of times the wafer area, so the wafer consumes precursor tens of times faster than the blank wafer the recipe was developed on. Third, nobody models these two things together: feature-scale conformality models [1, 2, 4, 8] assume the precursor density at the feature mouth is given, and chamber models assume the wafer's uptake is given. The quantity that matters, the gas density at the wafer surface while the wafer is eating it, falls between the two.

This paper closes that gap with a model small enough to run on a laptop and checked hard enough to trust. A chamber flow model carries, on every wafer face, a feature-scale model of a hole; the two exchange precursor every time step. The feature model is verified against an independent ballistic code and fitted to published measurements. The result is not a simulation to be rerun for every recipe but a rule: pulse time is set by precursor supply and patterned area, and by nothing else that a chamber recipe normally adjusts.

Equipment makers are building exactly this class of model under the name of equipment-and-process digital twins [15]. This work is an open, reproducible instance for ALD, with every case file, boundary condition and figure in the repository.

Contributions: (i) a two-way coupled chamber–feature wall condition for OpenFOAM, released; (ii) three-method verification (continuum, this work's voxel Monte Carlo, Argonne's Machball) and public-data validation of the feature model; (iii) the supply-limit rule and its range of validity; (iv) the blanket-to-product recipe-transfer factor, the shielding mechanism and the edge-lag inversion; (v) the temporal-versus-spatial design chart and the plasma-ALD recombination ceiling.

2 Model

2.1 Chamber

Axisymmetric showerhead chamber, 300 mm wafer, 15 mm gap, 200 mm chamber radius, annular outlet. N_2 carrier at 300 Pa, 573 K ($\rho = 1.76 \times 10^{-3}\,\mathrm{kg/m^3}$, $\nu = 1.65 \times 10^{-2}\,\mathrm{m^2/s}$), 300 sccm (0.050 m/s at the showerhead face), Re = 0.045, residence time 0.53 s. Steady laminar flow (`simpleFoam`, OpenFOAM v2606 [19]), frozen for the species step. TMA as a passive scalar c/c_0, $D = 8.4 \times 10^{-3}\,\mathrm{m^2/s}$ at 300 Pa (Chapman–Enskog scaling), gap Péclet number 0.09. Inlet: Danckwerts total-flux condition

$$U\,(c_{\mathrm{face}} - c_0) = D\,\frac{c_{\mathrm{face}} - c_{\mathrm{cell}}}{\delta}. \tag{1}$$

We show that a fixed-concentration inlet over-injects by 11× at this Péclet number (Section 3.1).

2.2 Wafer boundary condition (concurrent multiscale)

Flat surface: irreversible Langmuir,

$$\frac{d\theta}{dt} = \frac{\bar{v}}{4}\,s\,(1-\theta)\,\frac{c_s}{N_0}. \tag{2}$$

Holes: cylinder of diameter d and aspect ratio AR, Knudsen diffusivity $d\bar{v}/3$, quasi-steady gas [4] with sink $(\bar{v}/d)\,[\,s\,(1-\theta(x)) + r\,]$, coverage advanced per cell; mouth density equals the local wafer-surface gas density c_s. Net uptake per unit wafer area

$$K_{\mathrm{eff}}\,c_s = (1-\phi) \times \text{flat uptake} + \phi \times \text{hole-mouth Knudsen flux}, \tag{3}$$

ϕ the open-area fraction. Implemented as an OpenFOAM mixed condition with value fraction $K_{\text{eff}}/(K_{\text{eff}} + D/\delta)$. Parameters: $s = 5 \times 10^{-3}$ (TMA) and 10^{-4} (H_2O) [5], $N_0 = 4 \times 10^{18}\,\text{m}^{-2}$, $\bar{v}$ from kinetic theory at 573 K, $d = 100\,\text{nm}$, hole density $1.78 \times 10^{13}\,\text{m}^{-2}$ (180 nm hexagonal pitch, 50 % of the wafer patterned): open area 14 %, $A_{\text{wall}}/A_{\text{wafer}} = 11$, 28, 56 for AR 20, 50, 100.

2.3 Verification and validation

- Scaling tests on the feature solver: penetration depth $\sim \sqrt{\text{dose}}$; dose for 95 % step coverage $\sim \text{AR}^2$ in the diffusion-limited regime (ratio 4.0 for AR 20 $\rightarrow$ 40 at $s = 0.5$; drops to 3.2 at $s = 0.05$ where the front becomes reaction-limited); sticking coefficient recovered from a synthetic profile within 5 % (log).

- Cross-code check: Machball [3] (ballistic, cosine re-emission) vs this Knudsen-continuum solver, 100 nm holes at 0.3 Pa TMA, 573 K, with Machball's wall discretisation converged ($N_z = 8$ to 16 AR): 90 % bottom coverage at 1.035 vs 1.158 s (AR 20), 4.230 vs 4.476 s (AR 50), 15.93 vs 16.19 s (AR 100), 95 % at 1.20 vs 1.33, 4.45 vs 4.70 and 16.2 vs 16.5 s; the continuum solver is 11, 5.5 and 1.6 % fast against converged Machball and 11, 7 and 3 % fast against the voxel Monte Carlo solver of Section 2.6, which agrees with converged Machball within 2 %. An earlier draft of this paper quoted Machball at $N_z = 2$ AR (95 % times 1.29 / 4.46 / 15.6 s), which Section 2.6 shows is not converged; the values here supersede them.

- Embedded-BC check: with negligible hole density the boundary condition's hole-bottom coverage matches the standalone solver driven by the same surface density to 1 % at all times.

- Validation against the Zenodo 3974438 PillarHall Al_2O_3 profiles [10, 9] (TMA/H_2O, 300 °C, 500 cycles × 0.1 s, 500 nm lateral channel, eight profiles averaged): the slit model with channel narrowing fits the saturation front to an rms of 0.038 of the plateau (Fig. 1). The front position (AR 215 at half thickness) fixes the exposure and its width fixes s, but both depend on the slit's Knudsen diffusivity. With the hydraulic-diameter value $D_K = 2w\bar{v}/3$ the fit gives $s_{\text{TMA}} = 8.5 \times 10^{-3}$ (profile-to-profile factor 1.5) and 7.5 Pa s per pulse (75 Pa TMA for 0.1 s). That value of D_K is exact for a cylinder but not for parallel plates, where grazing flights make the free-molecular diffusivity grow as $(w\bar{v}/4)\ln(L/w)$ (Clausing; Berman [49]): the 1D slit model is 24 % slow against Machball's ballistic trench and the voxel solver at AR 50 and 61 % at AR 100, and with the factor $(3/8)\ln(L/w)$ on D_K it is within 5 % of Machball from AR 20 to 100 and reproduces the ballistic front width to about 10 % at AR 500. With that correction (a factor 2.33 at AR 500) the same data give $s_{\text{TMA}} = 1.96 \times 10^{-2}$ (sheet-to-sheet $\times/\div$ 1.46) and 3.23 Pa s per pulse (32.3 Pa), at the same rms (Fig. 2): the profile cannot distinguish the two transport models, it only re-labels the parameters, s by the factor D rose and the exposure by its inverse. Published values are 0.5–2×10^{-3} [5] (275 °C set-point, extracted with a continuum slit model of the hydraulic-diffusivity family), 4×10^{-3} [22] (SFG, 300 °C) and 5.72×10^{-3} [4] (300 °C). The corrected value lies farther from all of them than the uncorrected one; the measured front is sharper than a ballistic slit with the literature s would make it. The ideal-plate factor is an upper bound for the real cavity, which is 140–200 gaps wide and carries support pillars that interrupt the grazing flights (finite width alone removes 8–13 % of the factor), so s from this profile lies between 8.5×10^{-3} and 1.96×10^{-2} depending on the cut-off length of the cavity; the chamber runs use 5×10^{-3} and the patterned-wafer conclusions are supply-limited and insensitive to it (Section 4).

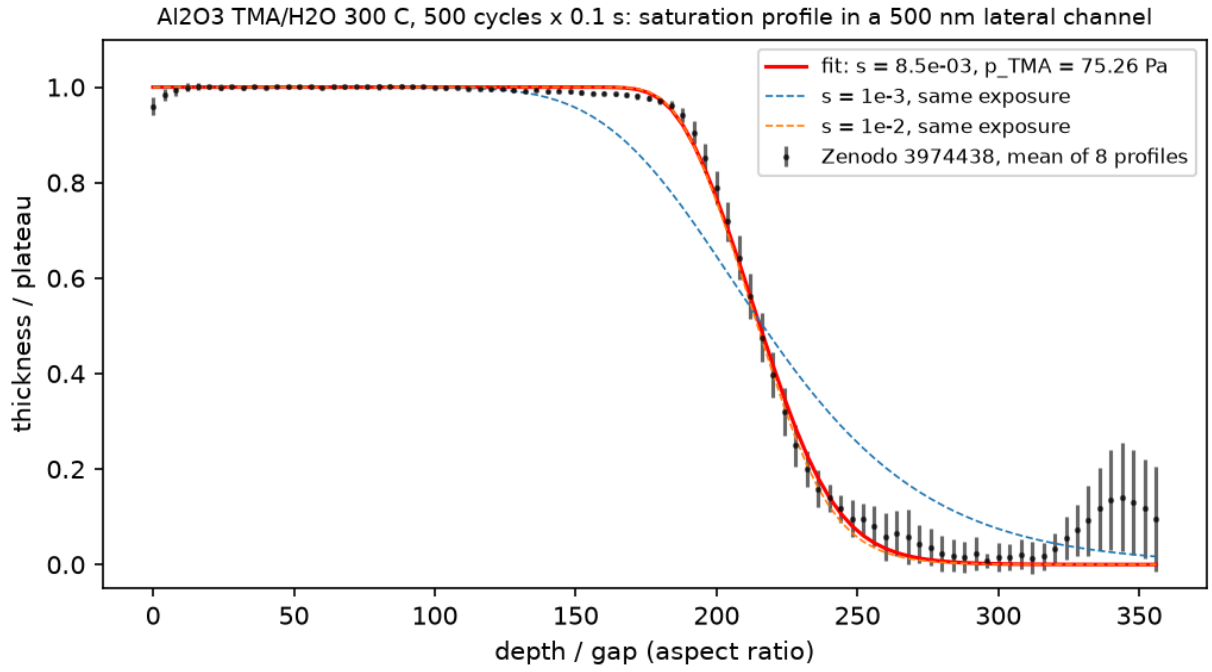


Figure 1: Validation of the feature model against the Zenodo 3974438 PillarHall data [10]: the mean of eight measured Al_2O_3 thickness profiles (TMA/H_2O, 300 °C, 500 cycles × 0.1 s) in a 500 nm lateral channel, normalised to the plateau thickness and plotted against depth expressed as aspect ratio. The red line is the two-parameter fit of the slit model with channel narrowing ($s_{\mathrm{TMA}} = 8.5 \times 10^{-3}$, 75 Pa TMA at the mouth); the dashed lines show $s = 10^{-3}$ and 10^{-2} at the same exposure.

2.4 Validation against a measured high-surface-area ALD run

A patterned wafer with tens of wafer-areas of holes is, to the chamber, a powder bed. The cleanest published measurement of supply-limited ALD is Argonne's fluidized-bed Al_2O_3 run on 940 m^2 of silica gel [12]: in-situ mass spectrometry put TMA breakthrough at 24 min, and the authors' own accounting gives 0.0115 ± 0.0020 mol of TMA supplied in that time against 0.0103 mol needed. In the normalisation used throughout this paper, saturation arrived at 1.12 (0.92 to 1.31) times the supply-limit time. Our twelve patterned-wafer cases, with nothing tuned to that experiment, saturate at 1.05 to 1.27 times theirs (Fig. 3); the blanket wafers, at a Damköhler number of 10 instead of hundreds, sit at 1.5 to 4.3 and are the kinetics-limited exception. Argonne's plug-flow theory [12, 13] predicts the same approach to unity from above. A second measured set, TMA utilization in a rotary-drum reactor at five silica-gel loadings from 40 to 1500 m^2 [14, Table 2], rises from 25 % to 71 % with area, i.e. saturation moves from 4.0 to 1.4 supply-limit times as the reactor fills with surface; our blanket and patterned wafers (0.07 to 4 m^2) and the fluidized bed (940 m^2, 92 %) fall on the same trend across five orders of magnitude of surface area (Fig. 3, right panel).

2.5 Chamber-scale checks

The blanket-wafer pulse of Section 3.1 can be compared with published saturation data once the exposure is defined with care. The recipe sets a feed exposure, 3 Pa of TMA for 0.345 s, 1.03 Pa s

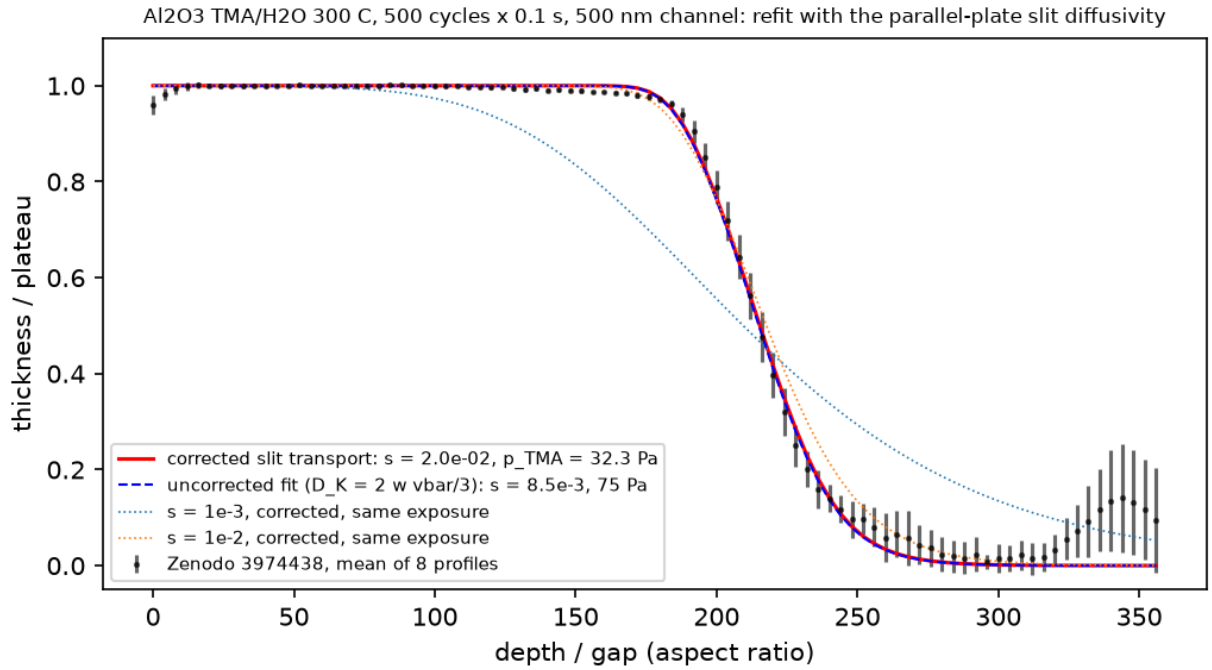


Figure 2: The same PillarHall profile refitted with the parallel-plate slit diffusivity, factor $(3/8)\ln(L/w) = 2.33$ on D_K at AR 500: $s_{\mathrm{TMA}} = 1.96 \times 10^{-2}$ and 32.3 Pa TMA at the mouth (red) lie on top of the hydraulic-diameter fit with 8.5×10^{-3} and 75 Pa (dashed blue) at the same rms of 0.038; the dotted curves show $s = 10^{-3}$ and 10^{-2} with the corrected transport at the same exposure.

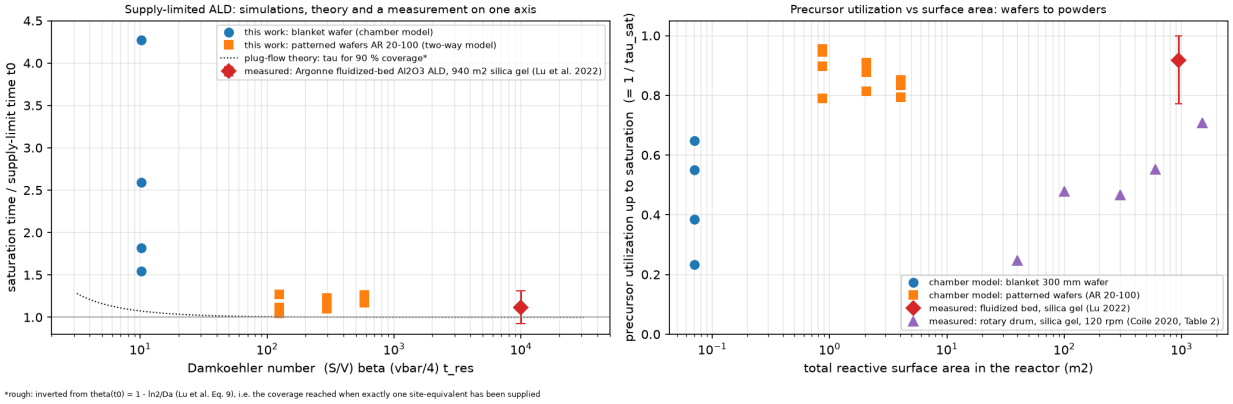


Figure 3: The supply-limit rule against measurements. Left: saturation time divided by the supply-limit time t_0 versus Damköhler number for the blanket and patterned chamber runs of this work, the plug-flow theory of Lu et al. [12], and the measured Argonne fluidized-bed Al_2O_3 run on 940 m^2 of silica gel. Right: precursor utilization up to saturation ($= 1/\tau_{\mathrm{sat}}$) versus total reactive surface area in the reactor, from a 0.07 m^2 blanket wafer to 1500 m^2 of powder, with the fluidized-bed point [12] and the rotary-drum loadings of Coile et al. [14].

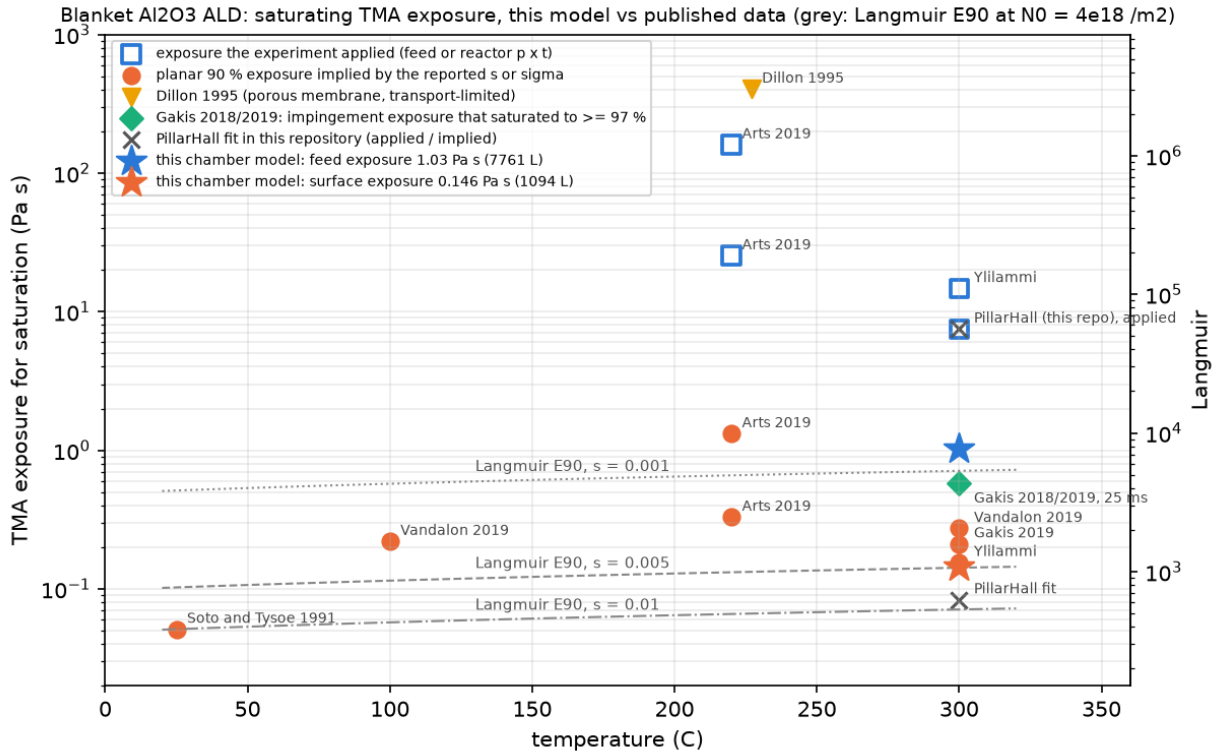


Figure 4: TMA exposure for saturation against temperature. Open squares: exposures the experiments applied (feed or reactor pressure times time); filled circles: planar 90 % exposure implied by each source's sticking coefficient or cross-section; diamond: the Gakis et al. 25 ms impingement exposure that gave at least 97 % saturation; crosses: the PillarHall fit of this work; stars: this chamber model, feed exposure (1.03 Pa s, 7,760 L) and surface exposure (0.146 Pa s, 1,090 L); grey lines: Langmuir E_{90} at $N_0 = 4 \times 10^{18}$ m^{-2} for $s = 10^{-3}$, 5×10^{-3} and 10^{-2}. Right axis in Langmuir.

or 7,800 Langmuir; the surface sees the time integral of its own TMA partial pressure, 0.146 Pa s or 1,100 Langmuir, because the wafer draws the gas down to 0.14 of the feed value on average while it saturates, and that surface exposure equals the Langmuir value $\ln(10)\, N_0/(sF_1)$ at $s = 5 \times 10^{-3}$, as it must. Seven independent sources give either the exposure they applied or a sticking coefficient or reaction cross-section from which the planar saturating exposure follows (Fig. 4, Table 1). The kinetic values span 0.05 to 1.3 Pa s: 0.27 Pa s from the sum-frequency cross-section of Vandalon and Kessels [24, 23], 0.21 from the reactor-fitted sticking coefficient of Gakis et al. [21], 0.16 from the PillarHall fit of Ylilammi et al. [4], 0.33 to 1.3 from Arts et al. [5] at 220 °C, 0.05 at room temperature in vacuum (Soto and Tysoe [26], as summarised by Puurunen [25]), 0.08 from our own PillarHall fit. Our 0.146 Pa s sits in the middle of the 300 °C cluster, within a factor 1.8 of each of the three sources that state a sticking coefficient at that temperature; the applied exposures (0.56 to 160 Pa s) are upper bounds that recipes choose to over-dose and say nothing sharper. Exposure scales as $1/s$ in this regime, so the model would reproduce the individual sources at s between 2.7×10^{-3} and 8.7×10^{-3}.

A second check uses the one published case that combines reactor-scale CFD with a measured thickness map in a commercial tool: the Ultratech Fiji F200 of Gakis et al. [20, 21] (200 mm wafer, 9.6 Pa Ar, 180 sccm, 300 °C, 0.63 mg of TMA per 25 ms valve opening, measured growth per cycle

Table 1: Blanket saturation exposures: this chamber model against published data. "Applied" exposures are what the experiment delivered (feed or reactor pressure times time); "kinetic E_{90}" is the planar 90 % exposure implied by the source's sticking coefficient or cross-section. Values read from figures are noted in `figures/blanket_saturation_literature.csv`.

Source	T (°C)	Kind of exposure	Exposure (Pa s)	Langmuir	s this model would need
This work, chamber model, centre: feed $p \times t_{90}$	300	applied	1.03	7,760	5×10^{-3} (as run)
This work, centre: surface integral of $p_s\,dt$ to $\theta = 0.9$	300	kinetic	0.146	1,090	5×10^{-3} (as run)
Ylilammi et al. 2018 [4], Picosun R-150, 300 Pa, 0.1 s pulse: fitted entrance pressure 147 Pa × 0.1 s (flat region saturated)	300	applied	14.7	110,000	upper bound only
same, implied by their fitted $\gamma = 5.72 \times 10^{-3}$ ($q = 5 \times 10^{18}$ m^{-2})	300	kinetic E_{90}	0.155	1,160	4.7×10^{-3}
Arts et al. 2019 [5], Oxford OpAL, set-point 275 °C: 190 mTorr s TMA dose, planar surface saturated	∼220	applied	25.3	190,000	upper bound only
same, 1200 mTorr s (H_2O-limited profile)	∼220	applied	160	1,200,000	upper bound only
same, implied by $s_0 = 2 \times 10^{-3}$ / 5×10^{-4} (Table I range; $N_0 = 4 \times 10^{18}$ assumed)	∼220	kinetic E_{90}	0.33 / 1.32	2,470 / 9,890	2.2×10^{-3} / 5.5×10^{-4}
Vandalon and Kessels 2019 [24], SFG, $\sigma_{TMA} = (6.5 \pm 0.6) \times 10^{-18}$ cm^2; temperature-independent 100–300 °C [23]	300 (100)	kinetic E_{90} from σ, no site density needed	0.273 (0.221)	2,050 (1,650)	2.7×10^{-3} (3.3×10^{-3})
Gakis et al. 2018 [20], Fiji F200, 9.6 Pa, 25 ms valve: wafer impingement 0.0118–0.0130 mol/m^2 (±0.0002) × 46.45 Pa s per mol/m^2; GPC 0.975–0.998 Å/cycle, i.e. ≥ 97 % saturated	300	applied (their CFD)	0.55–0.60	4,100–4,500	upper bound only
Gakis et al. 2019 [21], implied by fitted $s_0 = 0.004 \times 86$ % reaction probability, $N_0 = 4.1 \times 10^{18}$ from GPC 1.0 Å/cycle at 3500 kg/m^3	300	kinetic E_{90}	0.212	1,590	3.4×10^{-3}
PillarHall record (Zenodo 3974438 [10]), this work's fit: mouth exposure 75 Pa × 0.1 s	300	applied	7.5	56,000	upper bound only
same, implied by the fitted $s = 8.5 \times 10^{-3}$	300	kinetic E_{90}	0.084	630	8.7×10^{-3}
Soto and Tysoe 1991 [26], UHV, as summarised by Puurunen 2005 [25]: reaction probability 0.01	25	kinetic E_{90} ($N_0 = 4 \times 10^{18}$ assumed)	0.051	380	1.4×10^{-2}
Dillon et al. 1995 [27], porous alumina FTIR, as summarised by Puurunen 2005: $k_a = 0.0076$ s^{-1} at 0.01 Torr	227	first-order rate on a porous membrane: transport, not planar kinetics	404	3,000,000	(1.8×10^{-6}, not meaningful)

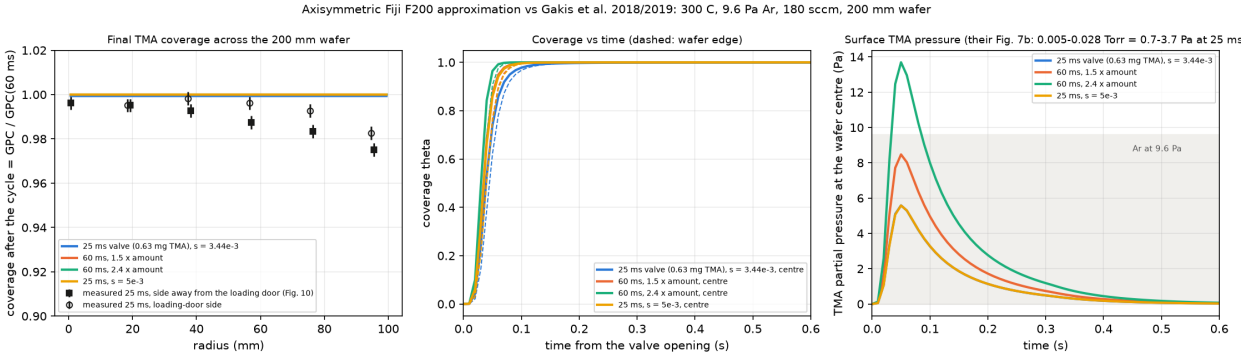


Figure 5: Axisymmetric approximation of the Fiji F200 against Gakis et al. [20, 21]. Left: coverage after the cycle across the 200 mm wafer for the four cases, with the measured 25 ms growth-per-cycle profile (their 2019 Fig. 10, digitised, divided by the 1.00 Å/cycle of the 60 ms pulse; filled squares the side away from the loading door, open circles the loading-door side, ±0.003). Middle: coverage against time at the centre (solid) and edge (dashed). Right: TMA partial pressure at the wafer centre; the shaded band is the 9.6 Pa of Ar.

0.975 to 0.998 Å/cycle along the diameter against 1.00 at 60 ms, non-uniformity 2.4 % falling to 0.58 %). We rebuilt the chamber as an axisymmetric wedge with dimensions read off their cross-section (the papers give none), a single axial inlet standing in for their three, the carrier flow frozen, their computed pulse amount and shape, and their kinetic model reduced to an irreversible sticking coefficient $s = 0.004 \times 0.86 = 3.4 \times 10^{-3}$ and $N_0 = 4.1 \times 10^{18}\,\mathrm{m}^{-2}$ from their growth per cycle (Fig. 5, Table 2). The model delivers 0.67 Pa s of TMA impingement to the wafer against 0.55 to 0.60 Pa s in their three-dimensional CFD and saturates the wafer with the 25 ms pulse (coverage 0.9994, 90 % at 70 ms) and completely with the 60 ms pulse, as the measurement does to within 2.5 %. It does not reproduce the 2.4 % non-uniformity: their exposure field is 10.7 % non-uniform because the pulse momentum through an off-axis inlet drives a recirculation over the wafer and the loading-door purge sweeps one side, ours is 0.3 % because an axisymmetric inlet into a diffusive gap smooths everything; and with irreversible Langmuir kinetics even their exposure variation would move coverage by under 0.5 %, so the remainder is their reversible, softer kinetics and the 21 °C temperature gradient and cold loading door they document. The chamber-scale transport that this paper's conclusions rest on, how much of the injected precursor the wafer takes before the pump does, is reproduced to the accuracy of the dimensions; the within-wafer uniformity of a particular injector is not a claim of this paper.

2.6 A 3D voxel Monte Carlo feature solver

To check the one-dimensional Knudsen-diffusion model against a description that makes none of its assumptions, the same features were solved with a test-particle Monte Carlo on a voxel grid (`lib/aldmc`). In the molecular-flow regime that holds inside the features (Kn $\sim$ 200), particles move in straight lines between wall collisions and never collide with each other. They enter through the feature mouth with a cosine angular distribution and a uniform position over the opening; on hitting a solid voxel face they stick with probability $s\,(1-\theta)$ on that face's own coverage, are lost with probability r, and otherwise re-emit diffusely; particles leaving through the mouth are lost. A time step launches M particles that represent the mouth flux $n\bar{v}/4$ times the mouth area and the step, and the coverage of every face is advanced by the Langmuir relaxation of the collisions it received,

Table 2: Fiji F200 reproduction against Gakis et al. [20, 21]: coverage after the cycle over $|r| \leq 95\,\mathrm{mm}$, within-wafer non-uniformity $(\max - \min)/\min$, time to 90 % after valve opening, TMA impingement exposure on the wafer, TMA captured by the wafer, and peak surface TMA partial pressure.

Case	θ after the cycle, centre / min / max	Non-uniformity	t_{90} centre / worst	Wafer exposure (Pa s)	Exposure non-uniformity	TMA captured	Peak surface TMA (Pa)
25 ms valve, 0.63 mg, $s = 3.44 \times 10^{-3}$	0.9994 / 0.9994 / 0.9994	0.0007 %	0.07 / 0.08 s	0.671–0.673	0.28 %	2.4 % of 5.26×10^{18} (41 monolayer-equivalents delivered)	5.6
60 ms, 1.5× amount	1.0000 / 1.0000 / 1.0000	0.00001 %	0.06 / 0.06 s	1.017–1.019	0.18 %	1.6 %	8.5
60 ms, 2.4× amount	1.0000 / 1.0000 / 1.0000	0.0000 %	0.05 / 0.05 s	1.639–1.640	0.11 %	1.0 %	13.7
25 ms, $s = 5 \times 10^{-3}$	1.0000 / 1.0000 / 1.0000	0.00003 %	0.06 / 0.07 s	0.671–0.673	0.28 %	2.4 %	5.6
Measured, 25 ms (2019 Fig. 10, digitised, / 1.00 Å per cycle at 60 ms)	0.996 / 0.975 / 0.998	2.37 % (2.40 % stated)					
Measured, 60 ms (stated)	1.00	0.58 %					
Their CFD + kinetics, 25 / 60 ms		2.45 % / 0.83 %		0.55–0.60 (2018 Fig. 12)	10.7 %		0.7–3.7 at 25 ms (2019 Fig. 7b)

the same update as the 1D solver with the Monte Carlo impingement in place of $n\bar{v}\,dt/4$. Because a voxel surface is a staircase with $4/\pi$ times the area of a smooth cylinder wall, each face carries sites and sticking in proportion to its projected area on the estimated smooth surface; without this correction the hole saturates 26 % late. The solver is deterministic for a given seed at any thread count, needs no diffusion coefficient or hydraulic diameter, and takes arbitrary shapes from an occupancy array (straight and tapered holes, slits, the word-line recess structure).

For the straight 100 nm holes at 0.3 Pa TMA and 573 K the voxel solver and Machball agree within 2 % at AR 20, 50 and 100 once Machball's wall discretisation is converged (t_{90} 1.15 / 4.51 / 16.45 s against 1.16 / 4.48 / 16.19 s; Machball's values move by 3–6 % between $N_z = 2\,\mathrm{AR}$ and 8 AR), while the 1D Knudsen solver is 11 % fast at AR 20, 7 % at AR 50 and 3 % at AR 100 (1.04 / 4.23 / 15.93 s): the continuum description converges to the ballistic one as the aspect ratio grows (Fig. 6, Table 3). The particle budget (25 000 to 400 000 per step), the voxel size (16 to 32 across the diameter) and the seed change t_{90} by at most 1 %; the time step enters linearly through the coverage frozen within a step (0.6 % at $dt = t_{90}/200$). The frustum with a 60 % bottom lengthens t_{90} by the factor 1.19 in the voxel solver and 1.20 in the 1D model. Driven by the chamber's centre-face density history of the long AR-100 case, the voxel hole reaches 90 % bottom coverage at 14.32 s (14.55 s with the history subsampled to 0.25 s) against 14.03 s in the 1D solver and 14.05 s in the boundary condition, with the same saturated and empty regions and a somewhat sharper front at 5, 10 and 13 s. For slits the two ballistic codes again agree (AR 50: 1.08–1.11 s voxel, 1.12 s Machball), but the 1D slit model with $D_K = 2w\bar{v}/3$ is 24 % slow, the known logarithmic correction of the parallel-plate Knudsen diffusivity; the word-line recess structure therefore saturates 27 % sooner in the voxel solver (10.7 s against 14.5 s at 0.3 Pa) with the same ordering of slit bottom and

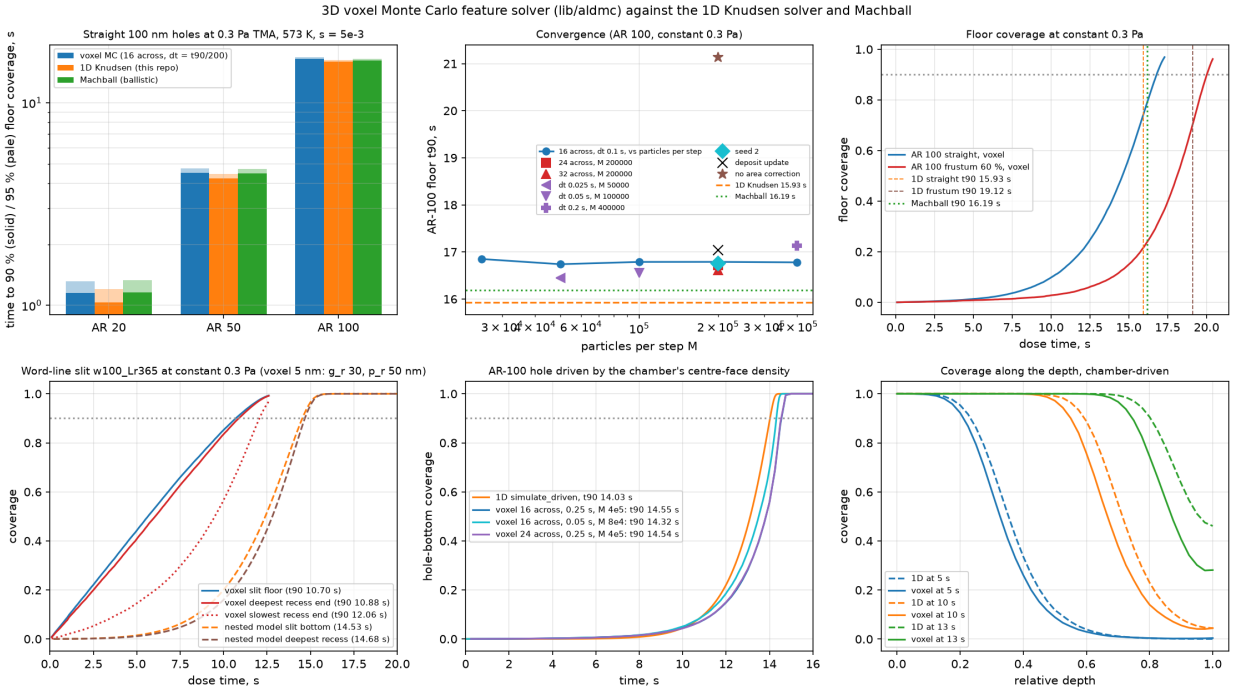


Figure 6: The voxel Monte Carlo feature solver against the 1D Knudsen solver and Machball. Bottom-coverage histories of straight 100 nm holes at AR 20, 50 and 100 (0.3 Pa TMA, 573 K, $s = 5 \times 10^{-3}$); convergence of the AR-100 t_{90} in particle budget, voxel resolution and time step; the tapered hole with a 60 % bottom; the word-line slit structure; and the AR-100 hole driven by the chamber's centre-face density history, with the coverage along the depth at 5, 10 and 13 s.

Table 3: Cross-check of the three feature solvers on straight 100 nm holes: time to 90 % and 95 % coverage of the hole floor at $s = 5 \times 10^{-3}$, $N_0 = 4 \times 10^{18}\,\mathrm{m}^{-2}$, 573 K, 0.3 Pa TMA. Voxel runs with 16 voxels across the diameter, $dt = 0.005$ / 0.0125 / 0.025 s, seed 1; Machball converged at $N_z = 8$ AR (16 AR at AR 100); the $N_z = 2$ AR column is the discretisation quoted in earlier drafts.

AR	Voxel t_{90} / t_{95} (s)	1D Knudsen t_{90} / t_{95} (s)	Machball converged t_{90} / t_{95} (s)	Machball $N_z = 2$ AR t_{90} / t_{95} (s)	Voxel vs Machball	Voxel vs 1D
20	1.148 / 1.316	1.035 / 1.204	1.158 / 1.334	1.123 / 1.289	−0.9 % / −1.3 %	+10.9 % / +9.3 %
50	4.513 / 4.735	4.230 / 4.451	4.476 / 4.704	4.247 / 4.465	+0.8 % / +0.7 %	+6.7 % / +6.4 %
100	16.454 / 16.784	15.931 / 16.249	16.191 / 16.511	15.279 / 15.590	+1.6 % / +1.7 %	+3.3 % / +3.3 %

recess ends. The chamber results rest on the hole model, whose Knudsen diffusivity $d\bar{v}/3$ is exact; the slit-based numbers of Section 3.11 are conservative by this factor (the parallel-plate correction of Section 2.3 removes it). The 24 voxel runs behind these numbers used 6.1×10^8 particles and 93 min of wall time in all on 7 to 8 threads, no single run above 10 min, against under a second per 1D solve and 1 to 131 s per converged Machball run.

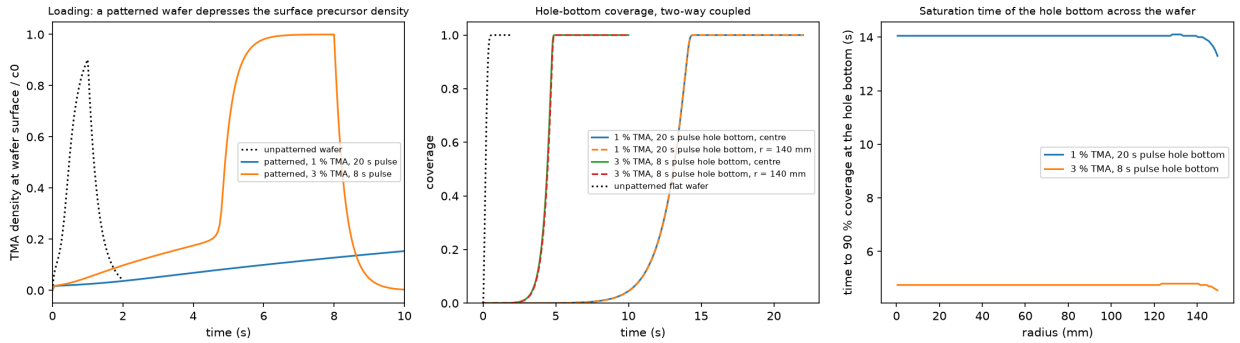


Figure 7: Two-way coupling on the 3D-NAND-like AR-100 wafer, 1 % TMA, 300 sccm, 20 s pulse. Left: TMA density at the wafer surface relative to c_0 for the unpatterned and the patterned wafer, showing the loading effect. Centre: hole-bottom coverage versus time at the wafer centre and at $r = 140$ mm, with the unpatterned flat wafer for comparison. Right: time to 90 % coverage at the hole bottom across the wafer radius.

3 Results

3.1 Blanket wafer

Fixed-concentration inlet absorbs 1179 % of the injected TMA; total-flux inlet 96.5 %. Centre flux plateaus at the supply rate Uc_0: at Pe = 0.09 the chamber is supply-limited. Dose uniform to 1 % out to $r = 100$ mm, 7 % dip at $r = 142$ mm (radial sweep), edge spike from the un-patterned floor ring. Gas phase clears 0.05 s after the pulse. Sticking sweep: capture 96 → 65 %, within-wafer spread 7 → 0.2 %, purge tail 0.15 → 0.58 s for $s = 1 \to 10^{-3}$. Self-limiting wafer, $s = 5 \times 10^{-3}$, 1 % TMA: 90 % coverage at 0.35 s (centre), 0.38 s (edge). Operating sweep: pulse time follows the supply time $N_0/(Uc_0)$ at 1.6 to 2.3×; pressure does not change it at fixed sccm but raises the edge lag from 5 % (100 Pa) to 14 % (1000 Pa); a 10 mm gap beats 25 mm on time and uniformity.

3.2 One-way coupling and the shielding effect

Driving a 100 nm hole with the blanket wafer's surface density: AR 50 bottom saturates at 0.92 s, against 4.45 s at constant density, because the flat surface stops consuming TMA at 0.35 s and the surface density rises tenfold. AR 100: 3 % coverage at the bottom after a 1 s pulse. Radius shifts the AR 50 time by only 10 %.

3.3 Two-way coupling: loading

3D-NAND-like wafer, AR 100, 1 % TMA: surface density collapses to $0.019\,c_0$ ($0.24\,c_0$ blanket), the wafer captures 94 % of the injected TMA during a 1 s pulse, and nothing saturates; a 1 s pulse delivers 8 % of the demand. Long pulses (Fig. 7): hole bottoms reach 90 % at 14.05 s (centre), 13.3 s (edge); flat regions between holes at 1.95 s (0.35 s blanket). 3 % TMA: 4.75 s.

3.4 The pulse-time map and the design rule

Table 4 gives the saturating pulse at the worst point on the wafer, and Fig. 8 plots it. Patterned wafers sit at 1.05 to 1.25× the supply limit $(1 + A_{\text{wall}}/A_{\text{wafer}})\, N_0/(Uc_0)$ over the whole map; blanket

Table 4: Pulse time to 90 % coverage at the worst point on the wafer, from sixteen two-way multiscale runs in the baseline chamber (300 Pa, 300 sccm, 15 mm gap; $s = 5 \times 10^{-3}$, $N_0 = 4 \times 10^{18}\,\mathrm{m}^{-2}$). Patterned wafers carry 100 nm holes at AR 20, 50 and 100, i.e. wall area 11, 28 and 56 times the wafer area.

TMA fraction	Blanket (s)	AR 20 bottoms (s)	AR 50 bottoms (s)	AR 100 bottoms (s)
0.5 %	0.65	5.37	13.4	28.1
1 %	0.38	2.71	6.73	14.1
3 %	0.18	0.95	2.32	4.79
10 %	0.09	0.33	0.75	1.51

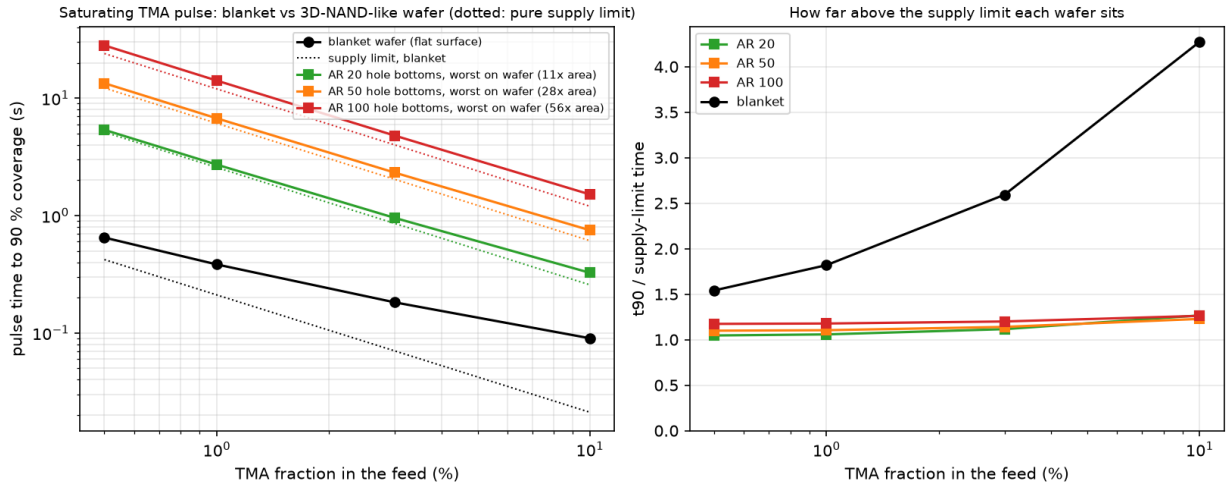


Figure 8: The pulse-time map. Left: saturating TMA pulse (time to 90 % coverage at the worst point on the wafer) versus TMA fraction in the feed for the blanket wafer and for 3D-NAND-like wafers with AR 20, 50 and 100 hole bottoms; the dotted line is the pure supply limit for the blanket wafer. Right: the same times divided by each wafer's own supply-limit time, showing the patterned wafers at 1.05–1.25 and the blanket wafer at 1.4–3.5.

wafers at 1.4 to 3.5× their own limit, rising with feed fraction as kinetics takes over. Design rule:

$$t_{\mathrm{pulse}} \approx 1.2\left(1 + A_{\mathrm{wall}}/A_{\mathrm{wafer}}\right)\frac{N_0}{U c_0}. \tag{4}$$

3.5 H_2O half-cycle

Blanket: 1.74 s (5× TMA; $s\bar{v}/4 = 0.02\,\mathrm{m/s} < U$). Patterned AR 100: 13.4 s (TMA 14.05 s), flat-to-bottom lag 2.6 s against 12 s for TMA. Cycle time ∼30 s at 1 % precursors vs ∼2.5 s blanket. The effect of reversible water adsorption on coverage and on the purge tail is shown in Fig. 9 and discussed in Section 4.

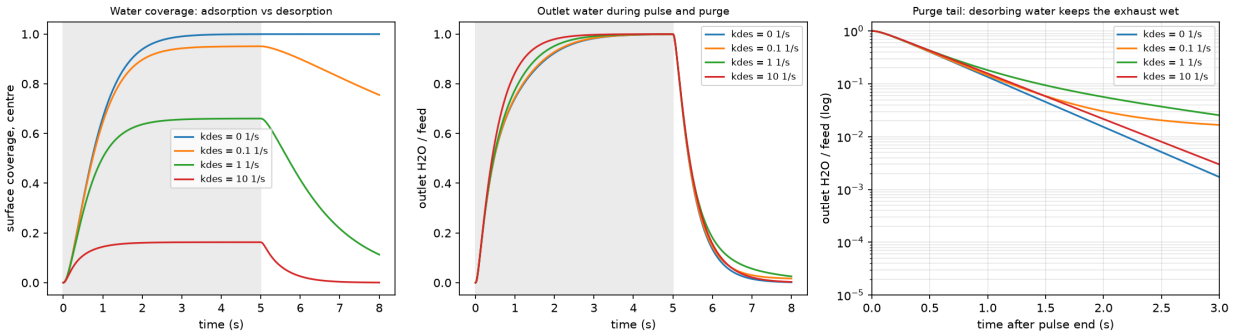


Figure 9: Reversible water adsorption on the blanket wafer, 1 % H_2O, 5 s pulse followed by a 3 s purge, for first-order desorption rates $k_{des} = 0$, 0.1, 1 and 10 s^{-1}. Left: surface coverage at the wafer centre. Centre: outlet H_2O relative to the feed during pulse and purge. Right: the purge tail on a logarithmic scale; desorbing water keeps the exhaust above 1 % of the pulse level for seconds at $k_{des} = 0.1$ to 1 s^{-1}.

3.6 Temporal versus spatial ALD

The design rule gives the molecules required per unit wafer area per half-cycle,

$$\Phi_{req} = 1.2\left(1 + A_{wall}/A_{wafer}\right) N_0. \tag{5}$$

A temporal showerhead tool delivers them at Uc_0 over the whole wafer and pays two purges per cycle; a separated-chemistry (spatial) tool delivers them in passes through a precursor zone at a flux k times the showerhead flux and pays rotation time instead of purges. With $k = 30$, a 1 s rotation and a 25 % zone, the two architectures tie on blanket wafers (2 s per cycle) and diverge linearly with patterned area: 6.8 vs 2 s at $A_{wall}/A_{wafer} = 11$ (AR 20), 29.9 vs 4 s at 56 (AR 100), 55.9 vs 8 s at 112. The crossover lies at A_{wall}/A_{wafer} of 5 to 10 for $k = 30$; k is the one quantity a tool maker knows and we do not, so Fig. 10 is a design chart in k. This is the quantitative form of the argument behind separated-chemistry ALD architectures for 3D NAND [16, 18].

3.7 Plasma ALD: the recombination ceiling

With a wall loss probability r for radicals, the time to 90 % bottom coverage in a 100 nm AR-100 hole rises from 8.0 s ($r = 0$) to 8.6 s (3×10^{-5}), 10.2 s (10^{-4}), 16 s (3×10^{-4}) and 64 s (10^{-3}), and the bottom is not reached at $r \geq 3 \times 10^{-3}$; the largest aspect ratio saturating within a 5 s exposure falls from 79 ($r = 10^{-5}$) to 56 (10^{-3}) and 29 (10^{-2}), Fig. 11. For O atoms on SiO_2 and TiO_2 ($r \sim 6 \times 10^{-5}$ [6, 7]) recombination is a 10 to 25 % penalty at AR 100; on high-recombination surfaces it is a hard ceiling that no pulse length removes, and radical flux (plasma source design) or surface chemistry become the levers. The radical sticking coefficient (0.05) is assumed.

3.8 Operating optimum and uncertainty

A face-centred central composite design (25 two-way runs; TMA fraction 0.5–3 %, carrier 150–600 sccm, pressure 100–1000 Pa, gap 10–25 mm) on the AR-100 wafer, fitted as a response surface in JMP and as a Gaussian-process surrogate in MATLAB and scikit-learn (leave-one-out R^2 0.999 for log pulse time, 0.988 for utilization), gives the second-order corrections to the rule (Fig. 12). Pulse time scales as $x^{-1.0}$, $Q^{-0.8}$ and $p^{-0.12}$; the gap is not significant. Precursor utilization up to

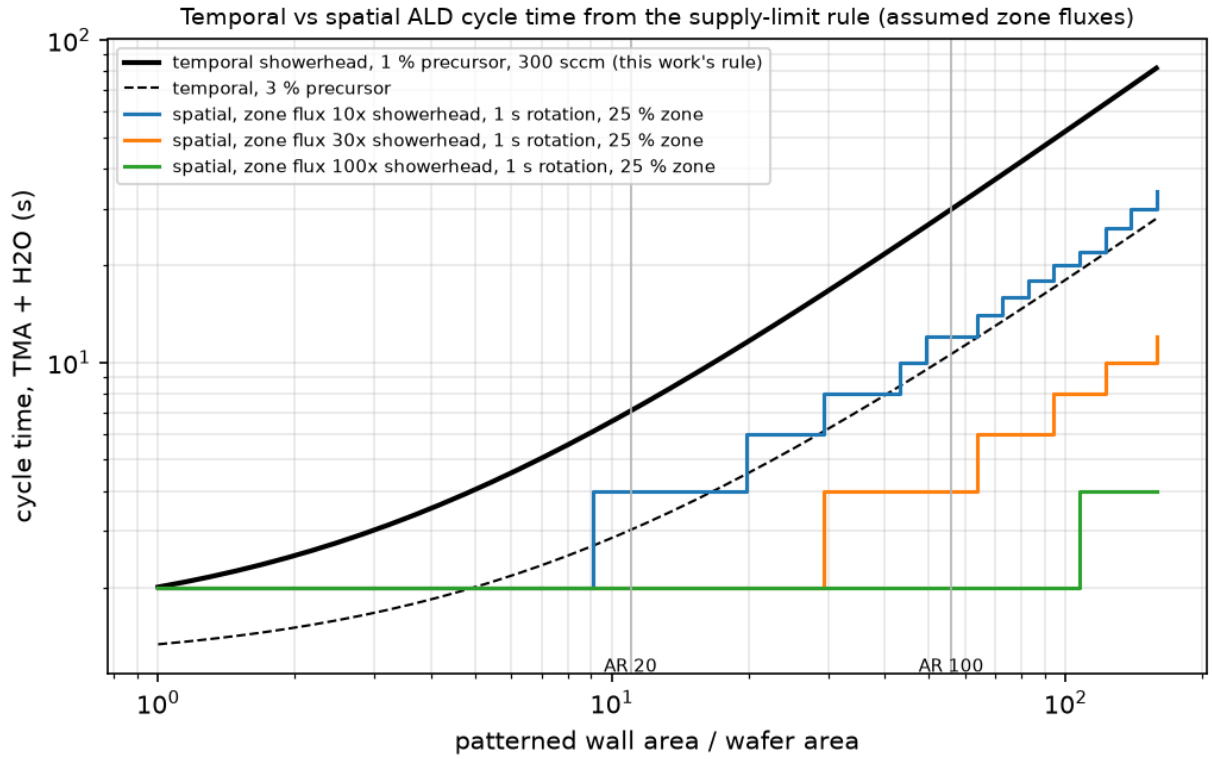


Figure 10: Temporal versus spatial ALD cycle time (TMA + H_2O) from the supply-limit rule, as a function of patterned wall area over wafer area. Solid and dashed black: temporal showerhead tool at 1 % and 3 % precursor, 300 sccm, with two purges per cycle. Steps: separated-chemistry (spatial) tool with a precursor-zone flux of 10, 30 and 100 times the showerhead flux, 1 s rotation and a 25 % zone; the zone fluxes are assumed, so the figure is a design chart in k. Vertical lines mark AR 20 and AR 100 of the 3D-NAND-like wafer.

saturation rises from 68 % at 100 Pa to 88 % at 1000 Pa and falls from 88 % at 150 sccm to 68 % at 600 sccm: carrier gas that is not needed to deliver precursor carries it past the wafer. The fastest point with edge lag under 5 % (a constraint never active: lag stays within −3.5 to +1.1 %) is 3 %, 600 sccm, 1000 Pa, 10 mm at 2.38 s. At fixed precursor flow, trading carrier gas for pressure (2.5 %, 150 sccm, 630 Pa, 10 mm) cuts the pulse from 11.6 to 9.8 s and raises utilization from 85 to 94 %. A Monte Carlo over the site density (lognormal, factor 1.25) puts a 90 % band of 8.2 to 17.2 s on the 11.9 s rule prediction at the design centre; the absolute pulse inherits the site-density uncertainty, the ratios in this paper do not. The scikit-learn GP reproduces the MATLAB leave-one-out R^2: 0.999 / 0.746 / 0.986 against 0.999 / 0.780 / 0.988 for time / edge lag / utilization. The cost-of-ownership front of pulse time against TMA per saturating pulse, over the factor ranges with edge lag under 5 % (Fig. 13): at the centre point's pulse time (11.6 s) the front uses 1.044 times the stoichiometric minimum of TMA against 1.181 at the centre, a 12 % saving per wafer pass at no cost in time; the last 1 % of saving (to 1.030 at 48.9 s) costs 4.2 times the pulse time.

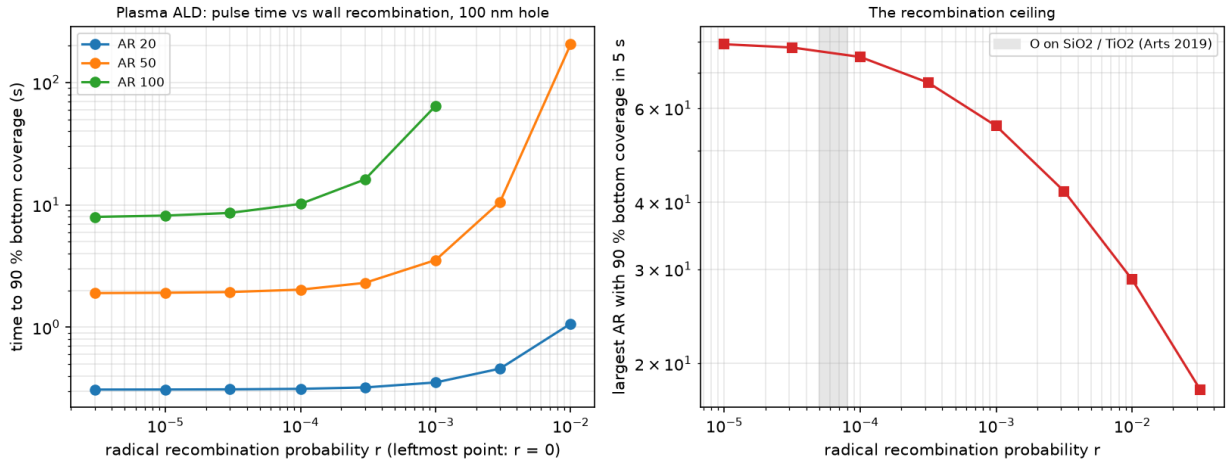


Figure 11: Plasma ALD in a 100 nm hole with radical sticking coefficient 0.05. Left: time to 90 % bottom coverage versus the radical recombination probability r on the hole walls for AR 20, 50 and 100 (leftmost point: $r = 0$). Right: the largest aspect ratio whose bottom reaches 90 % coverage within a 5 s exposure, versus r; the shaded band marks the measured range for O atoms on SiO_2 and TiO_2 [6].

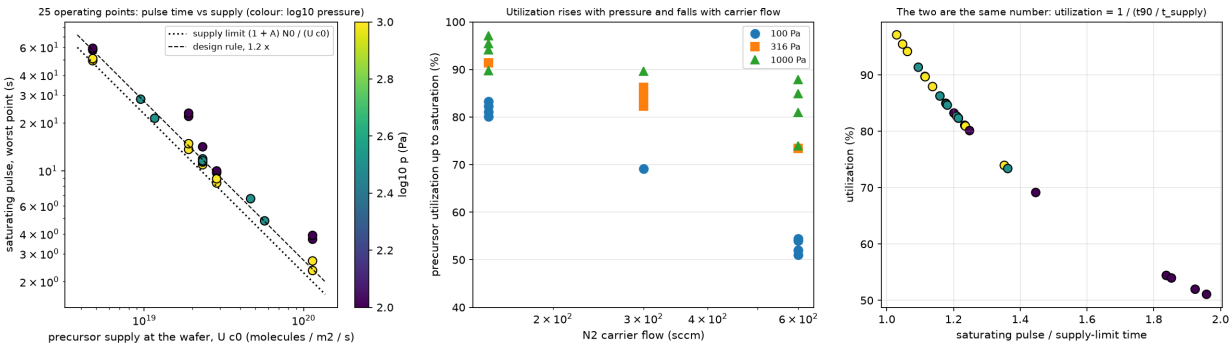


Figure 12: The 25-run central composite design on the AR-100 wafer. Left: saturating pulse at the worst point on the wafer versus precursor supply at the wafer Uc_0, coloured by $\log_{10}$ pressure, with the supply limit $(1 + A_{\text{wall}}/A_{\text{wafer}})\, N_0/(Uc_0)$ (dotted) and the 1.2× design rule (dashed). Centre: precursor utilization up to saturation versus N_2 carrier flow at 100, 316 and 1000 Pa. Right: utilization against the ratio of saturating pulse to supply-limit time, which are the same number.

3.9 Non-uniform layouts: the local wall-area ratio sets the pulse

The patterned-surface boundary condition takes hole density and aspect ratio as functions of radius (`ARProfile` and `holeDensityProfile` tables, interpolated at each face; a face with zero hole density is a flat Langmuir surface); the uniform case rerun with the rebuilt library is identical to the original to the 8 written digits at every face and write time. Two layouts at 1 % TMA, 300 sccm, 300 Pa (Fig. 14): (a) a dense AR-100 centre, full hole density for $r < 75$ mm ramping to bare wafer at 100 mm, wafer-average wall-area ratio 19.2; (b) full density everywhere with AR 100 at the centre ramping to 60 at the edge, wafer-average ratio 41.0. The rule with the wafer-average ratio under-predicts both: 5.1 s against the 11.85 s the dense centre needs, 10.6 s against 12.5 s for the AR ramp.

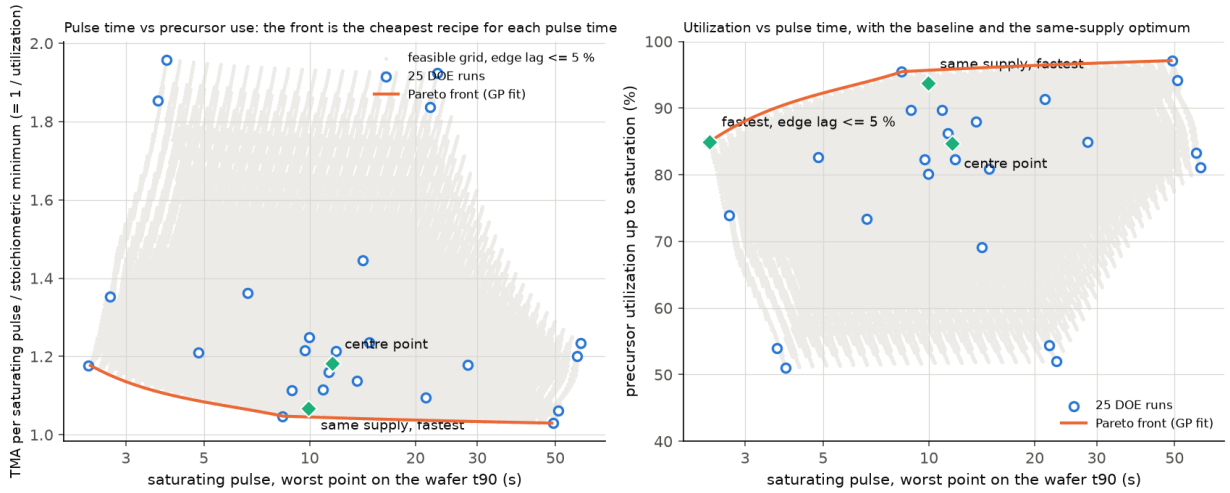


Figure 13: Cost-of-ownership front from the scikit-learn GP surrogate over the DOE factor ranges, restricted to points with edge lag $\leq 5\,\%$. Left: TMA delivered per saturating pulse relative to the stoichiometric minimum ($= 1/\text{utilization}$) versus the saturating pulse at the worst point on the wafer, with the 25 DOE runs, the centre point, the fastest feasible point and the fastest point at the centre's precursor supply. Right: the same front as precursor utilization versus pulse time.

With the local ratio of the most demanding region (55.9 at the centre of both) it predicts 14.4 s and is conservative by 15 to 21 %, against 2 % on the uniform wafer; the surplus is what the less-demanding neighbourhood feeds back laterally. A pattern boundary produces a saturation gradient inside the array: 11.85 s at the centre and 8.75 s at $r = 74.5$ mm, the last face at full density, a 26 % gradient where the uniform wafer varies by 0.4 %. The bare edge of layout (a) saturates 15 to 30 % slower than a blanket wafer (flat t_{90} 0.40 to 0.50 s against 0.35 to 0.38 s) because the array draws down the gas: surface TMA $0.163\,c_0$ over the bare region at 0.3 s against $0.24\,c_0$ on a blanket wafer. On the AR ramp the bottom time tracks the local aspect ratio (12.5 s at AR 100 to 7.6 s at AR 60) while the flat regions are unaffected (1.95 s): a 40 % depth variation gives a 39 % spread in saturation time, so the recipe is set by the deepest tier and shallower regions are over-dosed by their depth ratio.

3.10 Step coverage, purge and throughput

In fab units the rule reads as follows. Step coverage, the hole-bottom coverage over the flat top, follows the supply integral: at 1 % TMA the bottoms of AR 20, 50 and 100 holes reach 90 % at 2.7, 6.7 and 14.1 s (worst face), 95 % at 2.8, 6.8 and 14.2 s and 99 % at 2.9, 7.0 and 14.3 s (Fig. 15, Table 5). The front is sharp, $t_{99} - t_{90}$ is 0.25 to 0.3 s at every aspect ratio, so the rule's pulse (3.1, 7.3, 14.4 s) covers 99 % step coverage as well as 90 % (t_{99}/rule 0.95 to 1.00). The flat top between the holes saturates at 1.8 to 2.0 s regardless of aspect ratio. Converted to film at 0.103 nm per cycle (the PillarHall record's 51.5 nm after 500 cycles; the review value is about 0.1 nm per cycle [25]), 50 cycles of the blanket recipe (0.345 s) leave 1.4 nm of a 5.15 nm target on the top of an AR-100 array and nothing at the bottom, and a 20 % step coverage on an AR-20 array; 50 cycles of the rule's pulse leave 5.15 nm top and bottom on all three (Table 6).

The purge has three parts and only one of them is set by the chamber (Table 7). The hole itself empties by Knudsen diffusion in microseconds: for 100 nm $\times$ 10 μm and TMA at 573 K, $D_K = 1.4 \times 10^{-5}\,\mathrm{m^2/s}$, $L^2/D_K = 7\,\mu$s, and a transient solution with the mouth density set to zero

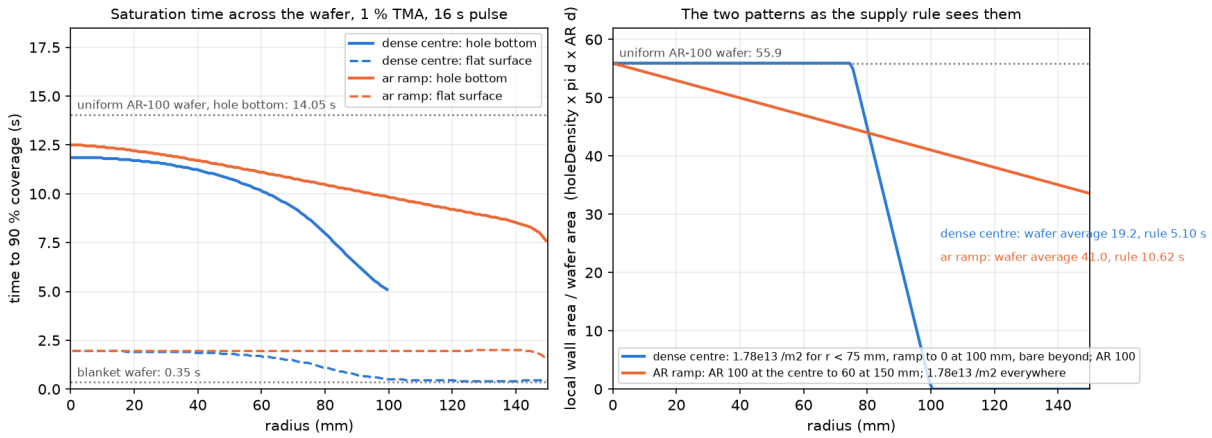


Figure 14: Two non-uniform layouts at 1 % TMA, 300 sccm, 300 Pa, 16 s pulse. Left: time to 90 % coverage at the hole bottom (solid) and on the flat surface (dashed) across the wafer for the dense-centre layout (AR 100, full hole density for $r < 75$ mm ramping to bare wafer at 100 mm) and the AR-ramp layout (full density, AR 100 at the centre to 60 at the edge), with the uniform AR-100 wafer (14.05 s) and the blanket wafer (0.35 s) as dotted references. Right: the local wall-area ratio of each layout versus radius, with the wafer averages (19.2 and 41.0) and the pulse the rule predicts from them (5.10 and 10.62 s).

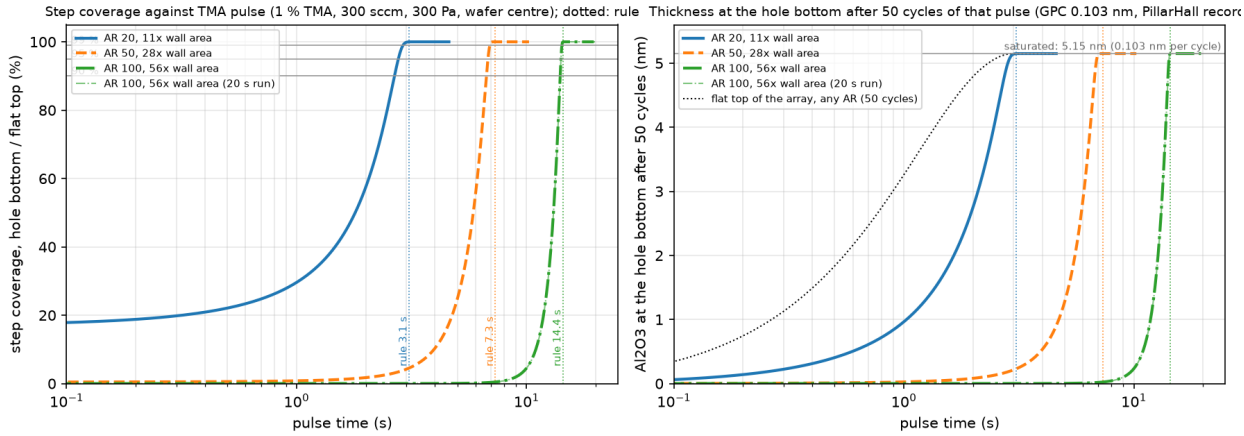


Figure 15: Left: step coverage (hole bottom over flat top, wafer centre) against pulse time for AR 20, 50 and 100 at 1 % TMA, with the 90, 95 and 99 % levels and the rule's pulse for each aspect ratio as dotted verticals (3.1, 7.3, 14.4 s); the 20 s AR-100 run is overlaid as the thin line. Right: Al_2O_3 thickness at the hole bottom after 50 cycles of that pulse (0.103 nm per cycle), with the flat top of the array as the dotted black line and the saturated 5.15 nm marked.

puts the bottom density at 1 % after 14 μs. The chamber outlet signal falls with a time constant of 0.36 to 0.39 s for TMA and 0.46 s for water, between the gap transit (0.3 s) and the chamber volume over the flow (0.53 s): a decade costs about 1 s, 1 % of the pulse-end level takes 1.9 s (TMA) and 2.2 s (water), 0.1 % about 2.7 and 3.2 s. Desorbing water is the long part: at $k_{\mathrm{des}} = 0.1\,\mathrm{s}^{-1}$ the exhaust is held at 1.6 % of the feed level by the surface 3 s after the pulse and reaches 1 % only after about 8 s; at $1\,\mathrm{s}^{-1}$, 4.3 s; at $10\,\mathrm{s}^{-1}$ the surface empties within 0.3 s and the purge is gas clearing

Table 5: Pulse for 90, 95 and 99 % bottom coverage at 1 % TMA, 300 sccm, 300 Pa, 15 mm gap (step coverage = $\theta_{\mathrm{bottom}}/\theta_{\mathrm{flat}}$); "map" rows are the pulse-time-map runs, the last row the 20 s AR-100 run with 0.05 s writes. The rule is $1.2\,(1 + A_{\mathrm{wall}}/A_{\mathrm{wafer}})\,N_0/(Uc_0)$.

Wafer (run)	$A_{\mathrm{wall}}/A_{\mathrm{wafer}}$	Rule (s)	t_{90} centre / worst (s)	t_{95} centre / worst (s)	t_{99} centre / worst (s)	Flat top t_{90} / t_{95} / t_{99} (s)	t_{99} / rule
AR 20 (map, 4.61 s pulse)	11.2	3.08	2.67 / 2.71	2.78 / 2.81	2.92 / 2.94	1.84 / 2.14 / 2.60	0.96
AR 50 (map, 10.26 s)	28.0	7.32	6.73 / 6.73	6.83 / 6.83	6.94 / 6.99	1.94 / 2.35 / 3.06	0.95
AR 100 (map, 19.68 s)	55.9	14.38	14.01 / 14.11	14.21 / 14.21	14.31 / 14.31	1.96 / 2.35 / 3.04	1.00
AR 100 (20 s run)	55.9	14.38	14.05 / 14.10	14.15 / 14.20	14.25 / 14.30	1.95 / 2.30 / 3.05	0.99

Table 6: Al_2O_3 after 50 cycles of a given pulse (0.103 nm per cycle; saturated film 5.15 nm), wafer centre: step coverage and bottom / top thickness.

Pulse	AR 20	AR 50	AR 100
0.345 s (blanket t_{90})	20 %, 0.26 / 1.29 nm	0.5 %, 0.01 / 1.32 nm	0 %, 0.00 / 1.44 nm
1 s	29 %, 0.95 / 3.23 nm	0.8 %, 0.03 / 3.25 nm	0 %, 0.00 / 3.16 nm
2 s	58 %, 2.76 / 4.79 nm	1.7 %, 0.08 / 4.69 nm	0 %, 0.00 / 4.67 nm
5 s	100 %, 5.15 / 5.15 nm	25 %, 1.27 / 5.15 nm	0.1 %, 0.00 / 5.15 nm
10 s	100 %, 5.15 / 5.15 nm	100 %, 5.15 / 5.15 nm	4.5 %, 0.23 / 5.15 nm
Rule (3.08 / 7.32 / 14.38 s)	100 %	100 %	100 %

again (2.4 s). k_{des} is a parameter, not a measurement, and the water purge in the cycle below is the $k_{\mathrm{des}} = 0.1\,\mathrm{s}^{-1}$ value by choice.

The full Al_2O_3 cycle, TMA pulse, purge to 1 %, water pulse, purge [50], is 12.2 s on the blanket wafer (0.35 + 1.9 + 1.7 + 8.3 s), 37.6 s on the AR-100 wafer (14.05 + 1.9 + 13.4 + 8.3 s) and 29.4 s at the same-supply optimum of Section 3.8 (9.8 + 1.9 + 9.4 + 8.3 s, the water pulse scaled by the TMA ratio since the water half-cycle was not in the design, and the purges held at the baseline values; the optimum's lower carrier flow raises the chamber residence time 2.8-fold, so its TMA purge would scale to about 5 s and its cycle to about 33 s, a scaling estimate rather than a run). At 0.103 nm per cycle, 5 nm is 49 cycles: 10 min per wafer and 5.5 wafers per hour with 60 s of handling on the blanket wafer, 31 min and 1.9 wafers per hour on the AR-100 wafer, 24 min and 2.4 at the optimum; 10 nm doubles the minutes (Fig. 16, Table 8). With gas clearing only for water ($k_{\mathrm{des}} = 0$, 2.2 s) the three cycles are 6.2, 31.6 and 23.3 s. The rule turns this into a throughput formula,

$$\mathrm{WPH} = \frac{3600}{N\,[\,2 \times 1.2\,(1 + A_{\mathrm{wall}}/A_{\mathrm{wafer}})\,N_0/(Uc_0) + \mathrm{purges}\,] + \mathrm{handling}}, \tag{6}$$

with N the number of cycles: below a wall-area ratio of about 10 the purges set the cycle and precursor fraction buys little; above it throughput falls as $1/(1 + A_{\mathrm{wall}}/A_{\mathrm{wafer}})$, and tripling the TMA fraction doubles the AR-100 throughput (1.8 to 3.5 wafers per hour) while leaving the blanket wafer where it was (5.9 to 6.3), Table 9. The chamber run at AR 100 sits on the 1 % curve within 2 %.

Table 7: Purge times. (a) The hole, 100 nm × 10 μm, TMA at 573 K ($\bar{v} = 410$ m/s, $D_K = d\bar{v}/3 = 1.37 \times 10^{-5}$ m^2/s), walls saturated, mouth density zero. (b) Chamber outlet signal after the pulse ends, as a fraction of the pulse-end outlet value; "extr." = extrapolated with the exponential tail fitted over the last 40 % of the written purge window. (c) Reversible water on the blanket wafer, 5 s pulse at 1 %, outlet to 1 % of the pulse-end value.

(a)

Quantity	Value
L^2/D_K	7.3 μs
Slowest eigenmode $4L^2/(\pi^2 D_K)$ (Dirichlet mouth, Neumann bottom)	3.0 μs
Bottom density to 1 %, first mode $(4/\pi)\exp(-t/\tau_1)$	14.4 μs
Numeric: bottom density to $1/e$	3.7 μs
Numeric: bottom density to 1 %	14.4 μs
Numeric: hole-mean density to 1 %	13.1 μs

(b)

Run	Pulse	Outlet at pulse end	To 10 %	To 1 %	To 0.1 %	Tail decay rate (time constant)
TMA, patterned AR 100, 20 s pulse, purge to 22 s	20 s	1.000 c_0	1.10 s	1.90 s	2.73 s (extr.)	2.75 s^{-1} (0.36 s)
TMA, blanket, 0.5 s pulse, purge to 1.5 s	0.5 s	0.297 c_0 (peak 0.367 at +0.11 s)	1.25 s (extr.)	2.14 s (extr.)	3.04 s (extr.)	2.57 s^{-1} (0.39 s)
TMA, blanket, 1.26 s pulse, purge to 2.26 s (map)	1.26 s	0.888 c_0	1.07 s (extr.)	1.95 s (extr.)	2.83 s (extr.)	2.63 s^{-1} (0.38 s)
H_2O, blanket, $k_{des} = 0$, 5 s pulse, purge to 8 s	5 s	0.999 c_0	1.14 s	2.20 s	3.25 s (extr.)	2.18 s^{-1} (0.46 s)
Reference: gap transit 0.015 m / 0.050 m/s				0.30 s		
Reference: chamber volume / flow, $\pi\,0.2^2 \times 0.015$ m^3 / 3.54×10^{-3} m^3/s				0.53 s		

(c)

k_{des} (s^{-1})	Coverage at pulse end / 3 s after	Outlet 3 s after the pulse	Purge to 1 %	Basis
0	1.000 / 1.000	0.0017 c_0	2.20 s	run (gas clearing)
0.1	0.951 / 0.755	0.0167 c_0 (tail model 0.0159)	8.26 s	extrapolated: tail $k_{des}\theta N_0 A/(Qc_0)$, θ decaying at 0.088 s^{-1}
1	0.660 / 0.113	0.0254 c_0 (tail model 0.0238)	4.33 s	extrapolated, θ decaying at 0.654 s^{-1}
10	0.163 / 0.001	0.0030 c_0	2.40 s	run (surface empties within 0.3 s; gas clearing again)

3.11 Tapered holes, word-line recesses and other chemistries

Real channel holes are not cylinders. The feature solver now takes a frustum (`TaperedHole`, diameter falling linearly from the mouth to a fraction of it at the bottom) and solves the conservative form $(1/A)\,\partial_x(AD\,\partial_x n)$ of the diffusion equation on the varying cross-section; with the sink switched off the density in a frustum is uniform to 10^{-9}, and a bottom ratio of 1 reproduces the cylinder exactly.

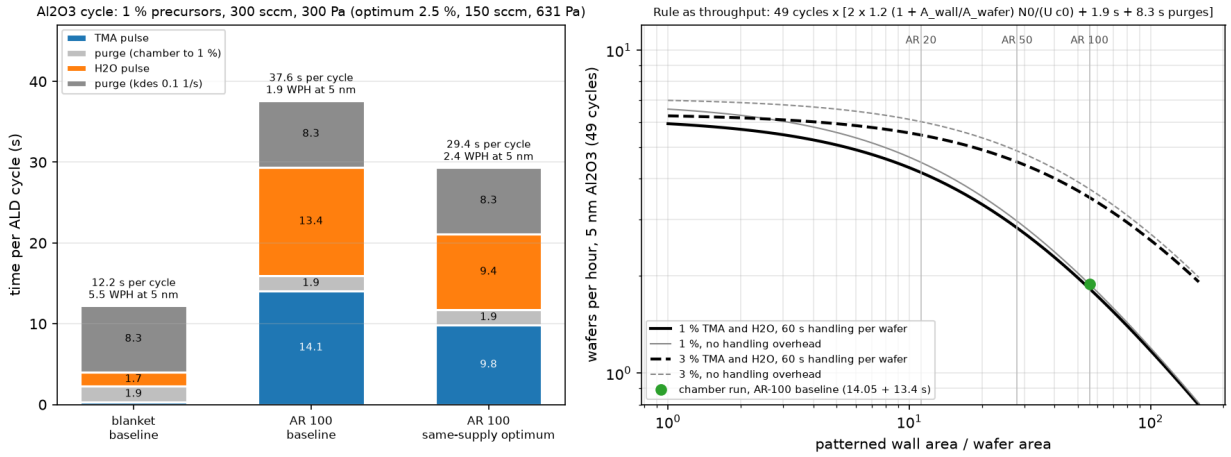


Figure 16: Left: the four segments of the Al_2O_3 cycle (TMA pulse, chamber purge to 1 %, H_2O pulse, purge at $k_{des} = 0.1\,s^{-1}$) stacked for the blanket baseline, the patterned AR-100 baseline and the same-supply optimum, with the cycle time and the 5 nm throughput (60 s handling) over each bar. Right: wafers per hour for 5 nm against wall-area ratio from the rule at 1 % and 3 % TMA, with and without 60 s handling, and the AR-100 chamber run as the point.

Table 8: Al_2O_3 cycle time and throughput (GPC 0.103 nm per cycle; 49 cycles for 5 nm, 98 for 10 nm). Purges: TMA 1.90 s (chamber outlet to 1 %, 20 s patterned run); H_2O 8.26 s ($k_{des} = 0.1\,s^{-1}$, a stated choice). In brackets: the same cycle with gas clearing only for water ($k_{des} = 0$, 2.20 s). WPH = wafers per hour.

Recipe	TMA + purge + H_2O + purge	Cycle	5 nm: min per wafer, WPH, WPH with 60 s handling	10 nm: min per wafer, WPH, WPH with 60 s handling
Blanket wafer, baseline (1 %, 300 sccm, 300 Pa, 15 mm)	0.345 + 1.90 + 1.74 + 8.26 s	12.2 s (6.2 s)	10.0 min, 6.0, 5.5 (5.1 min, 11.9, 9.9)	20.0 min, 3.0, 2.9 (10.1 min, 5.9, 5.4)
Patterned AR 100, baseline	14.05 + 1.90 + 13.4 + 8.26 s	37.6 s (31.6 s)	30.7 min, 1.95, 1.9 (25.8 min, 2.3, 2.2)	61.4 min, 0.98, 0.96 (51.5 min, 1.2, 1.1)
Patterned AR 100, same-supply optimum (2.51 %, 150 sccm, 631 Pa, 10 mm)	9.83 + 1.90 + 9.38 + 8.26 s	29.4 s (23.3 s)	24.0 min, 2.5, 2.4 (19.0 min, 3.2, 3.0)	48.0 min, 1.25, 1.2 (38.1 min, 1.6, 1.5)

Table 9: The rule as throughput: wafers per hour for 5 nm (49 cycles) against wall-area ratio, both half-cycles at the same supply flux, purges 1.90 + 8.26 s. Each cell gives cycle, WPH, WPH with 60 s handling, and in brackets the WPH with 60 s handling and the $k_{des} = 0$ water purge.

A_{wall}/A_{wafer}	1 % TMA	3 % TMA
1	11.2 s, 6.6, 5.9 (11.6)	10.5 s, 7.0, 6.3 (13.0)
11.2 (AR 20)	16.3 s, 4.5, 4.2 (6.4)	12.2 s, 6.0, 5.5 (10.0)
28.0 (AR 50)	25.1 s, 2.9, 2.8 (3.6)	15.1 s, 4.9, 4.5 (7.1)
55.9 (AR 100)	39.3 s, 1.9, 1.8 (2.1)	19.9 s, 3.7, 3.5 (4.9)
112	67.6 s, 1.1, 1.1 (1.2)	29.3 s, 2.5, 2.4 (3.0)

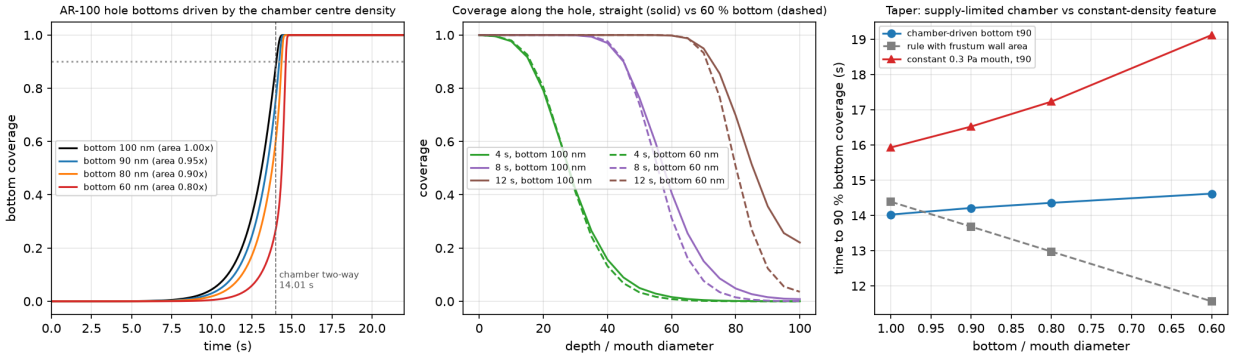


Figure 17: Tapered channel holes (AR 100, 100 nm mouth, 10 μm deep) with 100 / 90 / 80 / 60 nm bottoms. Left: bottom coverage against time driven by the chamber's centre-face TMA density (1 %, 20 s pulse), with the chamber's own two-way 14.01 s marked. Middle: coverage along the hole at 4, 8 and 12 s, straight (solid) against 60 % bottom (dashed). Right: time to 90 % bottom coverage against bottom/mouth diameter: chamber-driven, the rule with the frustum wall area, and the constant 0.3 Pa feature-only benchmark. Under the supply-limited chamber the taper costs 1–4 %; at constant mouth density 4–20 %; the rule with the reduced frustum area predicts 5–20 % less.

Published numbers for the taper of production holes are scarce: Lam reports less than 0.1 % CD deviation over 10 μm with cryogenic etching, about 10 nm on a 100 nm hole, against about 30 nm for room-temperature etches [38, 39], and TCAD studies use 0.3° tapers [37]; we therefore take 90 % as the anchored bottom diameter and 80 and 60 % as a sweep (no peer-reviewed vendor paper with a measured top/bottom CD pair in nanometres was found in open access). Driven by the chamber's centre-face TMA density of the AR-100 wafer, the bottoms of 100 / 90 / 80 / 60 nm-bottom holes reach 90 % at 14.03 / 14.21 / 14.36 / 14.62 s (Fig. 17, Table 10): the taper costs 1 to 4 %, because on a supply-limited wafer the hole's own conductance is not the bottleneck. At a constant mouth density the same taper costs 4 to 20 % (15.9 to 19.1 s): the lost conductance ($D_K \sim d$, $A \sim d^2$) outweighs the lost wall area. The rule with the frustum wall area (0.95 / 0.90 / 0.80 of the cylinder's) moves the other way, to 13.7 / 13.0 / 11.6 s, and the two bracket a tapered-hole wafer: the area reduction buys at most the rule's 5 to 20 %, the transport penalty takes back 1 to 4 % of it, and for the anchored 90 % taper both are within 5 % of the straight hole.

Word lines of a replacement-gate 3D-NAND stack are filled through the gate-line slit into lateral cavities a few tens of nanometres high and hundreds of nanometres to a micrometre deep. We model this as a nested feature (`aldconf.recess`): a slit of width w_s and depth L_s whose walls carry recesses of height g_r, length L_r and pitch p_r, each recess a Knudsen–Langmuir slit sub-problem fed by the local slit density, with the slit wall's uptake equal to the flat uptake on the un-recessed fraction plus the recess mouth flux on the recessed fraction, solved quasi-steadily; it is the `patternedSurface` construction one level down, and it reduces to the plain slit without recesses (to 10^{-9}) and to a slit with the same total wall area when the recesses are short (within 3 % at aspect ratio 1). With literature dimensions (29 nm word lines at 51 nm pitch [37], 98 pairs in 5 μm [28], 250 nm slits with recesses of 365 nm [41] or 1 μm [40]) and the TMA drive of the AR-100 wafer, the slit bottom saturates at 5.4 and 8.3 s and the last recess bottom 0.09 and 0.72 s later (2 and 9 %); a 100 nm slit takes 13.5 and 16.5 s with lags of 0.5 and 1 % (Fig. 18, Table 11). In every case the structure saturates within 4 % of the plain slit carrying the same total wall area (14.7 and 39.6 times the slit wall), so the rule's area-only treatment of recesses holds up to recess aspect ratios

Table 10: Tapered AR-100 holes (100 nm mouth, 10 μm deep), 1 % TMA, baseline chamber, driven one-way by the chamber's centre-face TMA density of the straight-hole wafer.

Bottom diameter	Wall area / straight	Chamber-driven bottom t_{90} (one-way)	Rule with frustum area, $1.2\,(1 + 55.9\,A_{\mathrm{rel}})\,N_0/(Uc_0)$	Driven / rule	Constant 0.3 Pa mouth, t_{90}
100 nm (straight)	1.000	14.03 s (chamber two-way: 14.05 s first write, 14.01 s interpolated)	14.39 s	0.97	15.93 s
90 nm (literature-anchored, cryogenic etch)	0.950	14.21 s	13.68 s	1.04	16.52 s
80 nm	0.900	14.36 s	12.98 s	1.11	17.23 s
60 nm	0.800	14.62 s	11.56 s	1.26	19.12 s

Table 11: Word-line recesses through a slit, TMA ($s = 5 \times 10^{-3}$, $N_0 = 4 \times 10^{18}\,\mathrm{m}^{-2}$), 5 μm stack, 29 nm recesses at 51 nm pitch, driven one-way by the AR-100 wafer's centre-face TMA density, and at a constant 0.3 Pa mouth density, with the hydraulic-diameter slit diffusivity (the parallel-plate-corrected values are given in the text). A_f is the total wall area over the plain slit wall. The effusion-limit exposure is the stoichiometric minimum at the mouth, $(A_{\mathrm{wall}}/A_{\mathrm{mouth}})\,N_0/(n_0\bar{v}/4)$.

Slit width w_s	Recess length L_r (AR_r)	Area factor A_f	Chamber-driven t_{90}: plain slit bottom	Slit bottom (flat wall)	Recess bottom at mouth / mid / stack bottom	Delay of the last recess bottom behind the slit bottom	Area-equivalent slit bottom	Constant 0.3 Pa: slit bottom / last recess bottom; exposure	Ratio to the effusion-limit exposure
250 nm	365 nm (12.6)	14.7	2.30 s	5.37 s	2.09 / 4.46 / 5.46 s	0.09 s (1.8 %)	5.39 s	2.70 / 2.79 s; 0.84 Pa s	4.6
250 nm	1000 nm (34.5)	39.6	2.30 s	8.33 s	2.94 / 6.94 / 9.05 s	0.72 s (8.6 %)	8.70 s	6.02 / 7.03 s; 2.11 Pa s	4.3
100 nm	365 nm (12.6)	14.7	3.76 s	13.46 s	2.11 / 8.55 / 13.53 s	0.07 s (0.5 %)	13.49 s	14.53 / 14.68 s; 4.40 Pa s	9.7
100 nm	1000 nm (34.5)	39.6	3.76 s	16.52 s	2.97 / 13.82 / 16.73 s	0.20 s (1.2 %)	16.66 s	37.2 / 39.2 s; 11.7 Pa s	9.6

of 35. The consequence for the recipe is in the area: a wafer tiled with these slits at a 1 to 2.3 μm pitch carries 150 to 189 wafer-areas of wall, 2.7 to 3.4× the AR-100 hole wafer, and the rule puts its TMA pulse at 38 to 48 s (fully patterned) at 1 %. Closing this with a recessed-slit boundary condition in the chamber is the natural next step; the nested model says the correction it would find inside the structure is below 10 %. The 1D slit transport behind these numbers (the hydraulic-diameter diffusivity, as in Table 11) was 24 % slow against the ballistic solvers (Section 2.6); with the parallel-plate correction of Section 2.3 the chamber-driven times become 5.1 / 8.0 s (250 nm slits) and 11.0 / 15.4 s (100 nm slits), the recess lags 0.1–0.5 s (1–7 %), the area-equivalent slit holds within 3 %, and the constant-density w100_Lr365 structure saturates in 10.0 / 10.2 s against the voxel solver's 10.7 / 10.9 s (6 % fast, the continuum-versus-ballistic offset of an AR-50 slit) instead of 14.5 / 14.7 s (36 % slow); the wall-area ratios and the rule are geometry only and unchanged.

The rule carries no chemistry beyond N_0 and c_0, and a second chamber chemistry tests that. HfO_2 from TDMAH ($M = 354.7$, $\bar{v} = 185$ m/s at 573 K) was run on the AR-100 wafer at 1 %, 300 Pa, 573 K, 300 sccm with $s = 7.0 \times 10^{-3}$ [28] (calibrated to 3D-NAND HfO_2 step coverage; no direct TDMAH measurement exists), $N_0 = 2.77 \times 10^{18}\,\mathrm{Hf/m^2}$ from a GPC of 0.10 nm [29, 30] and the bulk density 9.68 g/cm^3 [32], and a gas diffusivity of $3.6 \times 10^{-3}\,\mathrm{m^2/s}$ (hard-sphere Chapman–

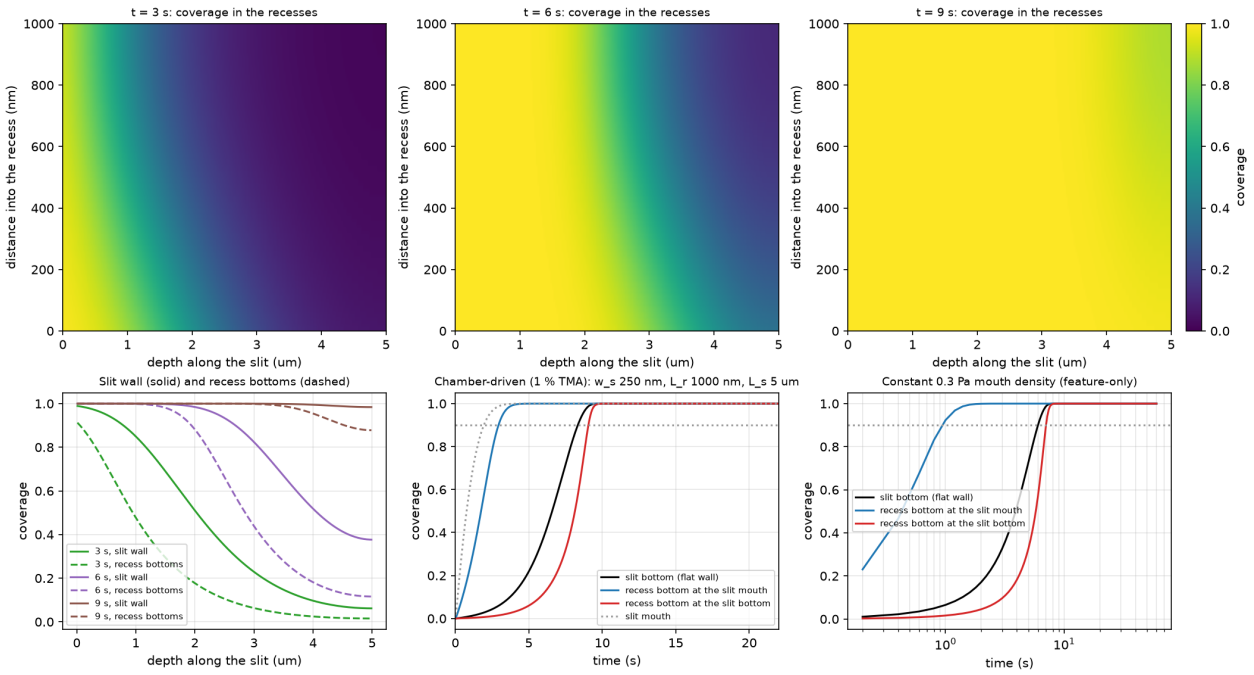


Figure 18: Word-line recesses fed through a slit (250 nm slit, 5 μm deep, 29 nm recesses 1 μm long at 51 nm pitch, TMA), chamber-driven. Top row: coverage in the recesses (x = depth along the slit, y = distance into the recess) at 3, 6 and 9 s. Bottom left: coverage of the slit wall (solid) and of the recess bottoms (dashed) along the slit at the same times. Bottom middle: slit mouth, slit bottom and recess bottoms at the mouth and at the stack bottom against time under the chamber drive. Bottom right: the same at a constant 0.3 Pa mouth density. The saturation front runs down the slit and into each recess behind it; the last recess bottom lags the slit bottom by 0.7 s (9 %) under the chamber drive, and the structure saturates within 4 % of a slit with the same total wall area.

Enskog relative to TMA). The hole bottoms reach 90 % at 11.6 s against 14.05 s for TMA; the rule gives 9.96 s against 14.38 s (Table 12). TDMAH sits 17 % above its rule where TMA sits 2 % below: 0.5 s of the 1.6 s excess is the 2.3× lower diffusivity (gap Péclet number 0.21 instead of 0.09; a run with the TMA diffusivity gives 11.1 s), the rest the shorter supply time against the fixed chamber overheads, the trend the DOE found for the rule constant. 573 K is at the upper edge of the TDMAH self-limiting window [29, 30, 31]; it was kept so that the flow field, U, c_0 and the inlet condition are those of every other run, and the rule at 250 °C is 9.3 to 10.0 s because Uc_0 at fixed sccm and fraction is temperature-independent. Rule-only predictions for TiN from $TiCl_4$ ($s = 0.10$ [4], GPC 0.30 Å, density 4.23 g/cm^3 [33], $N_0 = 1.24 \times 10^{18}$) and W from WF_6 ($s = 0.40$ [35], GPC 2.5 Å [36], 19.25 g/cm^3, $N_0 = 1.58 \times 10^{19}$) give 4.4 and 57 s per cycle on the same wafer at 1 %: the sticking coefficients span 5×10^{-3} to 0.4 and do not enter, since $s\bar{v}/4$ exceeds the inlet speed by 6 to 400× for all four precursors; what differs is the number of metal atoms a cycle has to deliver. Per nanometre of film the pulse scales with the film's metal-atom number density (HfO_2 100 s/nm, Al_2O_3 144, TiN 148, W 227), which is the number a tool's throughput model needs. The inputs carry their caveats: the TDMAH sticking value is a model calibration for an undisclosed Hf amide, not a measurement; the $TiCl_4$ value is for $TiCl_4/H_2O$ on TiO_2 at 110 °C, not $TiCl_4/NH_3$; the WF_6 value is on the Si-terminated surface, not the B-terminated one; the HfO_2 and W film densities are bulk values, not measured ALD densities; N_0 therefore carries the GPC × density uncertainty of

Table 12: Other chemistries. (a) Chamber runs on the AR-100 wafer (wall-area ratio 55.9), 1 % precursor, 300 Pa, 573 K, 300 sccm, 20 s pulse: TMA, TDMAH, and TDMAH with the TMA gas diffusivity as a sensitivity. (b) Rule-only predictions for the AR-100 wafer at 1 % precursor in the baseline chamber ($Uc_0 = 1.90 \times 10^{19}$ molecules m^{-2} s^{-1}); $N_0 = \mathrm{GPC}\,\rho N_A/M$.

(a)

	TMA (reference)	TDMAH	TDMAH with the TMA diffusivity (sensitivity)
M, $\bar{v}$ at 573 K	72.09, 410 m/s	354.7, 185 m/s	354.7, 185 m/s
s, N_0 (m^{-2})	5×10^{-3}, 4.0×10^{18}	7.0×10^{-3}, 2.77×10^{18}	7.0×10^{-3}, 2.77×10^{18}
D in N_2 at 300 Pa; inlet fraction f	8.4×10^{-3} m^2/s; 1.49×10^{-3}	3.62×10^{-3} m^2/s; 3.45×10^{-3}	8.4×10^{-3}; 1.49×10^{-3}
Gap Péclet number Ug/D	0.09	0.21	0.09
Flat t_{90} (centre)	1.95 s	1.30 s	1.20 s
Hole-bottom t_{90} centre / worst / $r = 149$ mm	14.05 / 14.10 / 13.50 s	11.60 / 11.60 / 11.10 s	11.10 / 11.15 / 10.95 s
Rule $1.2\,(1 + 55.9)\,N_0/(Uc_0)$	14.38 s	9.96 s	9.96 s
Bottom t_{90} / rule	0.98	1.17	1.12
Surface density at 5 s (centre)	$0.084\,c_0$	$0.197\,c_0$	$0.213\,c_0$

(b)

	Al_2O_3 / TMA	HfO_2 / TDMAH	HfO_2 / TDMAH	TiN / $TiCl_4$	W / WF_6
T	300 °C	300 °C	250 °C	325 °C	300 °C
s (source)	5×10^{-3} (chamber runs; Arts et al. [5] 0.5–2×10^{-3}, PillarHall fit 8.5×10^{-3})	7.0×10^{-3} (Reiter et al. [28], Hf amide, calibrated to 3D-NAND HfO_2 step coverage)	7.0×10^{-3}	0.10 (Ylilammi et al. [4], $TiCl_4/H_2O$ on TiO_2 at 110 °C; no $TiCl_4/NH_3$ value published)	0.40 (Elam et al. [35], WF_6 on the Si-terminated surface at 573 K; no B-terminated value)
GPC (source)	0.10 nm	0.10 nm (Blaschke et al. [29]: 1.08 Å at 200 °C, 0.93 Å at 250 °C; Gieraltowska et al. [30]: 1.2 Å at 180–240 °C)	0.093 nm (Blaschke et al. [29])	0.030 nm (Kinnunen et al. [33], 325 °C; Satta et al. [34]: 0.024 nm at 400 °C)	0.25 nm (Klaus et al. [36], WF_6/Si_2H_6)
Film density (source)	(N_0 taken as 4×10^{18}, one saturated monolayer)	9.68 g/cm^3 bulk monoclinic (Delabie et al. [32])	9.68 g/cm^3	4.23 g/cm^3 XRR (Kinnunen et al. [33]; bulk 5.21)	19.25 g/cm^3 (bcc, $a = 316.5$ pm)
N_0 (m^{-2})	4.0×10^{18}	2.77×10^{18}	2.58×10^{18}	1.24×10^{18}	1.58×10^{19}
$\bar{v}$	410 m/s	185 m/s	177 m/s	253 m/s	202 m/s
$s\bar{v}/4$ vs $U = 0.05$ m/s	0.51	0.32	0.31	6.3	20
Supply time $N_0/(Uc_0)$	0.211 s	0.146 s	0.136 s	0.065 s	0.830 s
Rule, AR-100 wafer	14.4 s	9.96 s	9.26 s	4.44 s	56.7 s
Rule, blanket (×1.2)	0.25 s	0.17 s	0.16 s	0.08 s	1.00 s

each chemistry (factor 1.1 to 1.3 across the cited values), the same dominant term the Monte Carlo on the TMA rule found. None of these moves the patterned-wafer result, which is supply-limited, but each would move a blanket-wafer one.

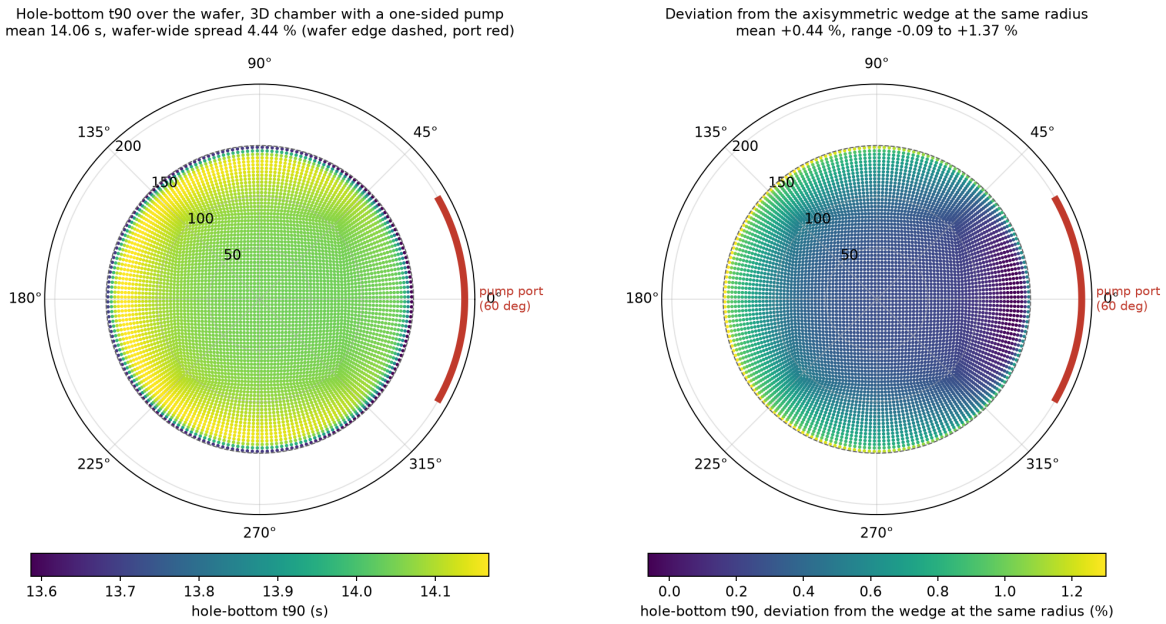


Figure 19: Full 3D chamber with a one-sided 60° pumping port: polar map of the hole-bottom t_{90} over the AR-100 wafer at the 6144 face centres (left, absolute; right, deviation from the wedge value at the same radius in %), the port marked in red at $r = 200$ mm, the wafer edge dashed.

3.12 A full 3D chamber with a one-sided pump

The wedge is axisymmetric by construction. To test what a real pumping arrangement does to the saturation map, the same chamber was meshed in full 3D (`blockMesh` O-grid, 106,000 hexahedra, non-orthogonality below 31°, 12 cells across the 15 mm gap) with the outlet reduced from the full annulus to a 60° sector of the side wall at $r = 200$ mm, the rest of the side wall a no-slip wall; showerhead, wafer condition, operating point and TMA fraction are those of Section 3.3 (uniform AR-100 wafer, 1 % TMA, 16 s pulse). The steady carrier flow (Re = 0.045) is strongly one-sided: the gap-averaged outward velocity at $r = 140$ mm is 0.85 m/s in the port-side quadrant and 0.01 m/s on the far side, against 0.27 m/s in the axisymmetric chamber, so gas over the far half of the wafer crosses the wafer toward the port instead of leaving radially (Fig. 19).

The gas phase sees the asymmetry, the saturation time hardly does. At $r = 140$ mm the surface TMA density is 9 % higher on the port side than on the far side at 0.3 s and 8 % higher at 14 s, and the hole-bottom coverage at 14 s is 0.90 against 0.84; but the hole bottoms fill in a steep late wave fed by the integrated supply, so this 6 % coverage lead is a 0.13 s lead in saturation time. The azimuthal peak-to-peak of the hole-bottom t_{90} is 0.09 % at $r = 50$ mm, 0.37 % at 100 mm and 0.96 % at 140 mm, in every case a single smooth lobe earliest at the port and latest opposite it (Fig. 20, Table 13). The slowest point on the wafer moves from the edge ring of the wedge to the far side at $r = 133$ mm, 160° from the port, at 14.17 s against 14.05 s in the wedge, and 0.3 % of that 0.8 % is the coarser 3D gap resolution, which shifts the centre value by the same amount. The flat surface between the holes, which saturates at 2 s while the gas phase is still drawn down to $0.02\,c_0$, is the more sensitive indicator: 1.92 s on the port side against 2.01 s on the far side at $r = 140$ mm, a 4.8 % spread.

The reason is the one behind the whole paper. At gap Péclet 0.09 the precursor reaches the

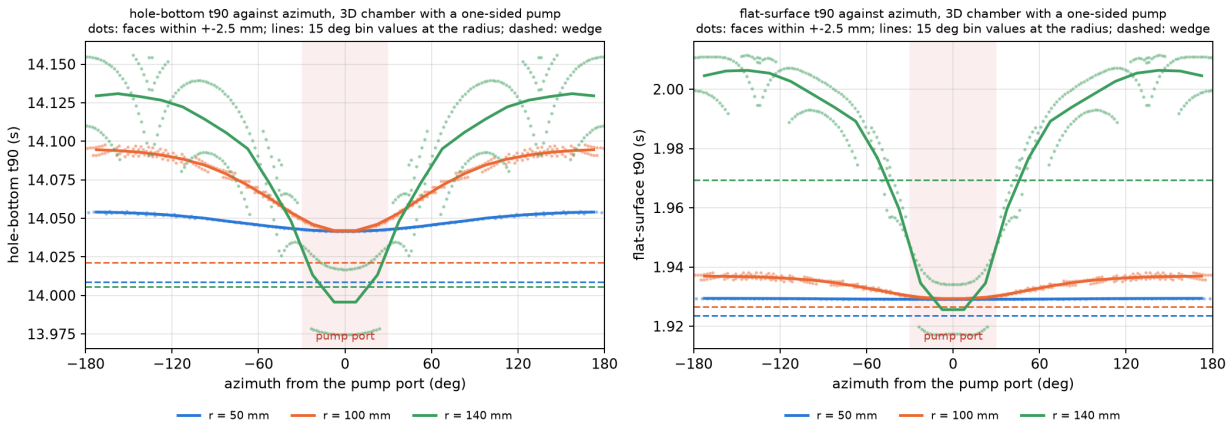


Figure 20: Hole-bottom (left) and flat-surface (right) t_{90} against azimuth from the port at $r = 50$, 100 and 140 mm in the 3D chamber: faces within ± 2.5 mm as dots, 15° bin values at the exact radius as lines, the wedge value at that radius dashed.

Table 13: Hole-bottom t_{90} in the 3D chamber by radius (0.9 crossing interpolated between 0.1 s writes; wedge values from the same interpolation on the 20 s AR-100 run; port at azimuth 0).

Radius (mm)	Faces	Mean (s)	Min (s)	Max (s)	Peak-to-peak, raw band (% of mean)	Peak-to-peak, 15° bins at the exact radius (%)	Port side − far side (%)	Wedge t_{90} at r (s)	3D mean / wedge
50	140	14.048	14.042	14.055	0.09	0.09	−0.08	14.009	1.0028
100	308	14.075	14.041	14.098	0.40	0.37	−0.36	14.021	1.0038
140	360	14.089	13.975	14.156	1.29	0.96	−0.90	14.006	1.0060
148.9 (outermost row)	248	13.687	13.544	13.931	2.83	–	−0.70	13.280 (r = 149.5 mm)	1.031

wafer by diffusion across the gap from the showerhead directly above each point, the patterned wafer captures almost all of what arrives, and the saturation time is the time to deliver $(1 + A_{\text{wall}}/A_{\text{wafer}})\, N_0$ per unit area at the showerhead's own rate. The cross-flow toward the port can redistribute only the few per cent that escape capture, and it does so over a lateral diffusion time ($R^2/D = 2.7$ s) short against the 14 s pulse. A one-sided pump therefore costs about 1 % of pulse time on this wafer; the recipe margin of the design rule (the factor 1.2) covers it, and the azimuthal lag would need a convective gap (Péclet of order 1, i.e. ten times the flow or pressure) before it mattered. The same ordering holds in the wedge's radial direction, where the edge lag is 0.4 %.

3.13 Byproducts: what a second species does and does not do

At the pressures of this work the mean free path is 12 to 123 μm, so inside a 100 nm feature the Knudsen number is 120 to 1,200 and molecules collide with the walls, not with each other: a reaction byproduct cannot block the precursor in the gas phase, and each species diffuses independently with its own Knudsen diffusivity (methane 2.1× faster than TMA, HCl 2.3× faster than $TiCl_4$). What a byproduct can do is surface-side: re-adsorb on a free site and block it until it desorbs, or etch the film. We extended the feature solver to two species (`aldconf.multispecies` and the library option): the byproduct is released at ν molecules per adsorbed precursor, solved quasi-steadily with

a source on the right-hand side, re-adsorbs on free sites with sticking s_B, desorbs at k_B and removes a coated site with probability k_{etch} per collision; the precursor sees the free fraction $1 - \theta_P - \theta_B$. With $s_B = k_{\text{etch}} = 0$, or $k_B \to \infty$, the single-species solver is reproduced to round-off, and the byproduct balance (produced + desorbed = escaped + re-adsorbed + etched) closes to 10^{-14} in every run. For TMA/H_2O the byproduct is methane (3 per TMA over a cycle [25]), which does not re-adsorb at these temperatures; the two-species result is the single-species one, and even a hypothetical re-adsorption at $s_B = 10^{-3}$ with 1 ms residence changes the AR-100 pulse by 0.01 % while 99.95 % of the methane leaves the hole (Fig. 21a, Table 14a). Every result of this paper therefore stands as computed.

For $TiCl_4/NH_3$ the byproduct is HCl, 4 per $TiCl_4$ over a cycle [42], about 0.5 of them in the $TiCl_4$ half-step [43], and its re-adsorption is documented on TiN, silica and TiO_2 [43, 44, 45]; no sticking coefficient or desorption rate exists for HCl on TiN at 573 K, so we ran the range that the HCl/alumina surface science spans (s_B 10^{-3} to 0.4 [47, 48]; k_B 0.1 to 1,000 s^{-1}, the 300–650 K TPD span of [47] read at 573 K). The desorption rate decides (Fig. 21b, Table 14b): at $k_B \geq 100\,s^{-1}$ HCl is a spectator whatever its sticking; at $k_B = 1\,s^{-1}$ the anchored half-step stoichiometry lengthens the 100 nm × 10 μm hole's saturating pulse by 11 to 55 % (7.75 s to 8.6–12.0 s at a constant 0.3 Pa mouth) and by 11 to 36 % under the chamber's density history (Table 15c); at $k_B = 0.1\,s^{-1}$ the blocked sites outlive the pulse and the hole fills three times slower. Releasing the whole cycle's four HCl in the half-step (a bound) doubles these costs, and an etch probability of 10^{-3} per HCl collision (a bound; HCl does not etch TiO_2 [45], and the TiN kinetic Monte Carlo of Kim et al. [46] lists Ti removal by surface Cl without a rate) costs 70 to 80 % at $\nu = 0.5$ and prevents saturation at $\nu = 4$: the hole settles into an etch-limited steady state. HCl accumulated in the chamber at 10 % of the precursor density adds 4 to 5 % at $k_B = 1\,s^{-1}$ and nothing at $100\,s^{-1}$, because the HCl produced inside the feature outweighs what the chamber brings in (Fig. 21d, Table 15d). The profiles at the rule pulse (Fig. 21c) show the mechanism: blocked sites near the mouth let the precursor pass, so the front is broader and deeper at a given time, but the sites behind it must be released and refilled, and that is where the time goes.

What this means for a chloride recipe is a measurement, not a number: the supply rule (Section 3.4) is untouched when $k_B \geq 100\,s^{-1}$, carries a 10 to 50 % chemistry margin at $k_B \sim 1\,s^{-1}$, and fails at $k_B \leq 0.1\,s^{-1}$ or with etching, so a TiN or HfO_2 tool model needs the HCl desorption rate on the growing film at the deposition temperature (one TPD of HCl from ALD TiN would fix it) before the wall-area rule can be applied with the same confidence as for Al_2O_3. The library carries the option behind defaults that reproduce the single-species boundary conditions bit for bit; a transported HCl field in the chamber is the next step.

4 Discussion

- Throughput on patterned wafers is bought with precursor delivery rate (vapour pressure, bubbler carrier flow, injector design), not with chamber pressure, gap or carrier flow; those knobs move uniformity.
- Recipe transfer: the blanket-to-product factor is the wall-area ratio, 11 to 56 here; monitor-wafer recipes under-dose product wafers by that factor at the same supply.
- Precursor utilization: 60 % (TMA) and 50 % (H_2O) over the saturating pulse on the patterned wafer; the remainder is pumped after saturation. Shorter pulses with feedback (end-point on surface density) would recover it.

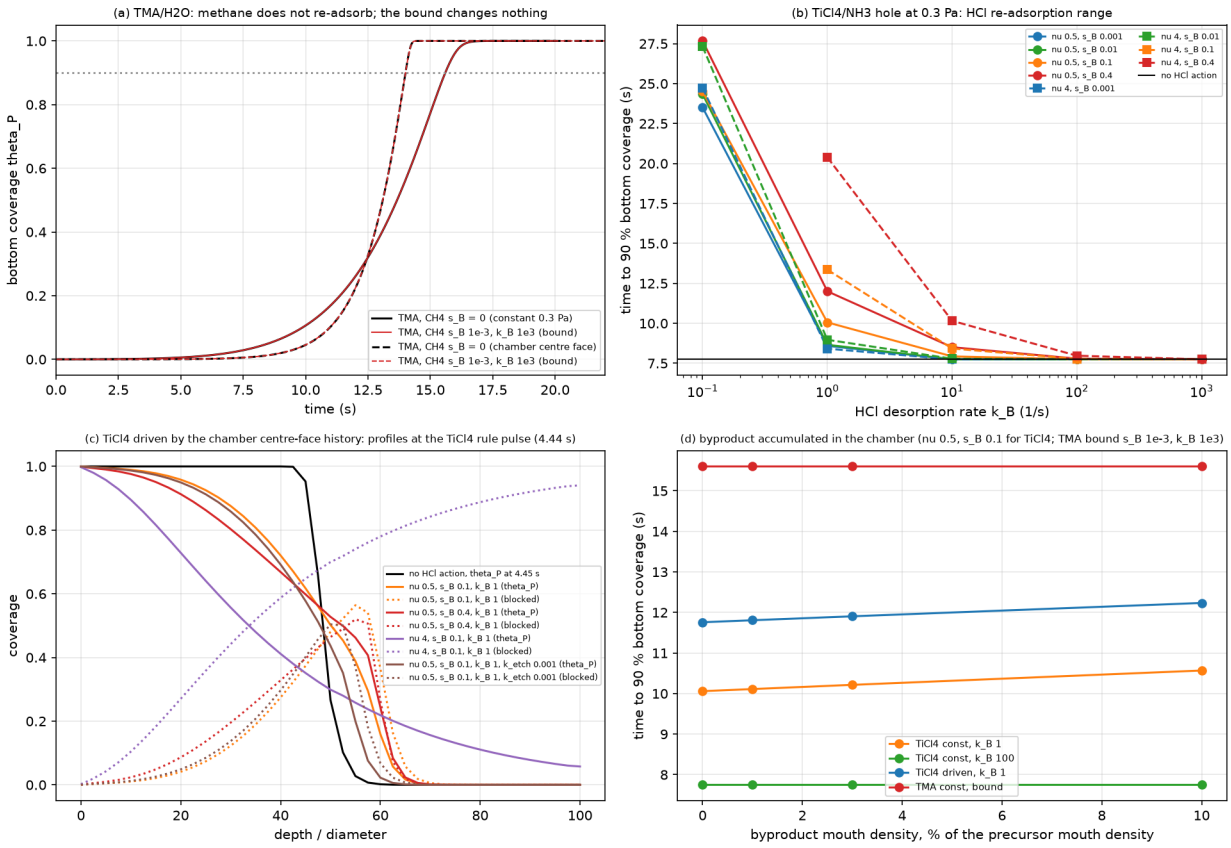


Figure 21: Two-species feature kinetics. (a) TMA bottom-coverage histories, single-species against the two-species model with methane at $s_B = 0$ and at the hypothetical bound, constant mouth (solid) and chamber-driven (dashed): the curves lie on top of each other. (b) $TiCl_4/NH_3$ hole at 0.3 Pa: time to 90 % bottom coverage against the HCl desorption rate k_B for s_B 10^{-3} to 0.4 and $\nu = 0.5$ (circles) and 4 (squares); the line is the single-species 7.75 s. (c) Coverage along the hole at the $TiCl_4$ rule pulse (4.44 s) under the chamber's centre-face history: precursor coverage (solid) and blocked coverage (dotted) for no HCl action, $\nu = 0.5$ at $s_B = 0.1$ and 0.4, $\nu = 4$ at $s_B = 0.1$ and the etch bound. (d) Time to 90 % against the byproduct mouth density (0 to 10 % of the precursor's) for the $TiCl_4$ cases at $k_B = 1$ and 100 s^{-1}, driven and constant, and the TMA bound.

- The TMA sticking coefficient is the least certain input of the chamber model: the published values at or near this temperature span 0.5–2×10^{-3} (Arts et al. [5], lateral-high-aspect-ratio front slope at a 275 °C set-point), $(4 \pm 1) \times 10^{-3}$ (Vandalon and Kessels [22], sum-frequency generation at 300 °C), 5.72×10^{-3} (Ylilammi et al. [4], fit to a lateral channel profile at 300 °C) and 8.5×10^{-3} to 1.96×10^{-2} (this work, fit to the Zenodo 3974438 profiles [10] at 300 °C, factor 1.5 sheet-to-sheet; the lower value with the hydraulic-diameter slit diffusivity of the literature extractions, the upper with the parallel-plate diffusivity that matches the ballistic solvers, Section 2.3). We reran the baseline operating point (1 % TMA, 300 sccm, 300 Pa, 573 K) at $s = 5 \times 10^{-4}$, 10^{-3}, 2×10^{-3}, 5×10^{-3} and 8.5×10^{-3} for the blanket wafer and the AR-100 patterned wafer (Fig. 22). On the patterned wafer the hole bottoms reach 90 % coverage at 14.00–14.15 s (centre) and 14.05–14.20 s (worst face) for every value, 1.17–1.18 times the supply-limit time of 11.99 s, with 84–85 % precursor utilization: a seventeen-fold

Table 14: Two-species results, 100 nm × 10 μm hole, part 1. (a) TMA/H_2O with methane (methane does not re-adsorb; bottom t_{90}). (b) $TiCl_4/NH_3$ with HCl at a constant 0.3 Pa mouth density, bottom t_{90} in s; without HCl action 7.754 s; upper block blocking only (ν, s_B, k_B), lower block with etching (k_{etch} per HCl collision on a Ti site, and the state at 30 s where the bottom never reaches 90 %).

(a)

Drive	Single-species	Two-species, CH_4 $\nu = 3$, $s_B = 0$	Bound: $s_B = 10^{-3}$, $k_B = 10^3\,s^{-1}$	Methane escaping the hole (bound)
Constant 0.3 Pa mouth	15.600 s	15.600 s, max coverage difference 0	15.601 s (+0.01 %)	99.95 %
Chamber centre-face history	14.005 s	14.005 s, max coverage difference 0	14.006 s (+0.01 %)	99.95 %

(b)

ν	s_B	$k_B = 0.1\,s^{-1}$	1	10	100	1,000
0.5 (anchored)	10^{-3}	23.51 (+203 %)	8.59 (+11 %)	7.75	7.75	7.75
0.5	10^{-2}	24.34 (+214 %)	8.65 (+12 %)	7.75	7.75	7.75
0.5	0.1	24.51 (+216 %)	10.06 (+30 %)	7.93 (+2 %)	7.75	7.75
0.5	0.4	27.67 (+257 %)	12.00 (+55 %)	8.51 (+10 %)	7.79 (+0.5 %)	7.75
4 (bound)	10^{-3}	24.73	8.41 (+8 %)	7.75	7.75	7.75
4	10^{-2}	27.32	8.97 (+16 %)	7.78	7.75	7.75
4	0.1	not within 30 s	13.36 (+72 %)	8.42 (+9 %)	7.77	7.75
4	0.4	not within 30 s	20.39 (+163 %)	10.16 (+31 %)	7.97 (+3 %)	7.75

ν	s_B	k_B (s^{-1})	$k_{etch} = 10^{-4}$	$k_{etch} = 10^{-3}$	Blocking only
0.5	10^{-2}	1	10.44	13.98	8.65
0.5	10^{-2}	100	9.32	12.46	7.75
0.5	0.1	1	11.44	14.17	10.06
0.5	0.1	100	9.74	13.11	7.75
4	10^{-2}	1	never: θ_P 0.045, blocked 0.94	never: 0.001 / 0.98	8.97
4	10^{-2}	100	never: 0.000 / 0.37	never: 0.000 / 0.37	7.75
4	0.1	1	never: 0.161 / 0.84	never: 0.011 / 0.99	13.36
4	0.1	100	never: 0.000 / 0.85	never: 0.000 / 0.85	7.77

change in s moves the saturating pulse by three write intervals. Low sticking changes the order of events inside the hole (at 10 s the bottom coverage is 0.38 at $s = 5 \times 10^{-4}$ and 0.02 at 8.5×10^{-3}) but not the finish, because the last tenth of the bottom coverage waits for molecules that the showerhead has not yet delivered. The design rule, the blanket-to-product transfer factor and the pulse-time map are therefore independent of which published value is right.

- The blanket wafer is not: low sticking is a conformality asset and a throughput cost only on blanket wafers. Its saturation time runs from 0.33 s at $s = 8.5 \times 10^{-3}$ to 1.01 s at 5×10^{-4}, about $s^{-0.4}$ rather than $1/s$, because the TMA density at the wafer rises as the uptake slows ($0.065\,c_0$ against $0.23\,c_0$ at the centre 0.1 s into the pulse) and partly compensates; the flat surface between the holes of the patterned wafer behaves the same way (1.55 to 6.00 s). Every blanket-wafer time in this paper therefore carries a factor-3 uncertainty from the sticking coefficient, whereas the patterned-wafer times carry only the site-density uncertainty of the supply limit. The 8.5×10^{-3} of our PillarHall fit is a slope-constrained value: the front slope fixes s and the penetration depth fixes the exposure per site, so the unknown TMA partial pressure trades against the assumed site density, not against s. It sits a factor 1.5 to 2 above the two other 300 °C values and a factor 4 above the upper bound of the 275 °C set-point data, a spread that also reflects three different kinetic definitions (initial sticking probability,

Table 15: Two-species results, part 2. (c) The saturating pulse chamber-driven (the TMA run's centre-face history as a stand-in; no $TiCl_4$ chamber run exists, so only the changes are meaningful) and at a constant 0.3 Pa mouth; without HCl action 9.622 s driven, 7.754 s constant. (d) Byproduct at the mouth (chamber accumulation, a parameter): bottom t_{90} for a byproduct mouth density of 0 / 1 / 3 / 10 % of the precursor's.

(c)

ν	s_B	k_B (s^{-1})	Driven bottom t_{90}	Change	Bottom θ_P at the $TiCl_4$ rule pulse 4.44 s	At the TMA rule pulse 14.4 s	Constant-mouth t_{90}	Change
0.5	10^{-2}	1	10.72 s	+11 %	0.000	0.997	8.65 s	+12 %
0.5	10^{-2}	100	9.62 s	0	0.000	1.000	7.75 s	0
0.5	0.1	1	11.76 s	+22 %	0.000	0.991	10.06 s	+30 %
0.5	0.1	100	9.62 s	0	0.000	1.000	7.75 s	0
0.5	0.4	1	13.11 s	+36 %	0.000	0.959	12.00 s	+55 %
0.5	0.4	100	9.67 s	+0.5 %	0.000	1.000	7.79 s	+0.5 %
4	10^{-2}	1	10.89 s	+13 %	0.012	0.997	8.97 s	+16 %
4	10^{-2}	100	9.62 s	0	0.000	1.000	7.78 s	+0.4 %
4	0.1	1	14.06 s	+46 %	0.058	0.922	13.36 s	+72 %
4	0.1	100	9.65 s	+0.3 %	0.000	1.000	7.77 s	+0.3 %
4	0.4	1	16.72 s	+74 %	0.118	0.698	20.39 s	+163 %
4	0.4	100	9.84 s	+2.3 %	0.000	1.000	7.97 s	+3 %
0.5, $k_{etch} = 10^{-4}$	0.1	1	12.51 s	+30 %	–	–	11.44 s	+48 %
0.5, $k_{etch} = 10^{-3}$	0.1	1	13.74 s	+43 %	–	–	14.17 s	+83 %

(d)

Case	0 %	1 %	3 %	10 %
$TiCl_4$ constant 0.3 Pa, $\nu = 0.5$, $s_B = 0.1$, $k_B = 1\,s^{-1}$	10.059 s	10.111 (+0.5 %)	10.215 (+1.6 %)	10.568 (+5.1 %)
$TiCl_4$ constant, $\nu = 0.5$, $s_B = 0.1$, $k_B = 100\,s^{-1}$	7.749 s	7.749	7.749	7.749
$TiCl_4$ chamber-driven, $\nu = 0.5$, $s_B = 0.1$, $k_B = 1\,s^{-1}$	11.759 s	11.808 (+0.4 %)	11.905 (+1.2 %)	12.234 (+4.0 %)
TMA constant, CH_4 bound $s_B = 10^{-3}$, $k_B = 10^3\,s^{-1}$	15.601 s	15.601	15.601	15.601

lumped Langmuir coefficient with an equilibrium constant, irreversible first-order Langmuir on 4×10^{18} sites). Correcting the slit transport does not close that gap: the 1D slit diffusivity was 24 % low at AR 50 and a factor 2.3 low at the channel's AR 500, and with the exact ballistic front width the same measured profile needs a larger s, not a smaller one. A profile grown at a known TMA partial pressure at 300 °C, with the site density taken from the measured growth per cycle, would remove the trade-off and fix s to the sheet-to-sheet scatter.

- Byproducts: at Knudsen numbers of 100 to 1,000 a byproduct cannot block the precursor in the gas, only on the surface. For TMA/H_2O (methane) the single-species model is exact to the purposes of this paper; for chlorides the HCl desorption rate at the deposition temperature decides between no effect ($k_B \geq 100\,s^{-1}$), a 10 to 50 % longer pulse ($k_B \sim 1\,s^{-1}$) and failure of the supply rule ($k_B \leq 0.1\,s^{-1}$ or etching), and that rate is not in the literature for TiN or HfO_2 at 573 K; the two-species solver and the library option (`aldkin` 1.1) are ready for it.

- Robustness checks: replacing the bare floor by a 25 mm edge ring with pumping below the wafer plane changes no saturation time by more than one write interval and leaves the within-wafer spread within 3 % (blanket 9–11 %, patterned 4 %), Fig. 23; the pulse-time results do

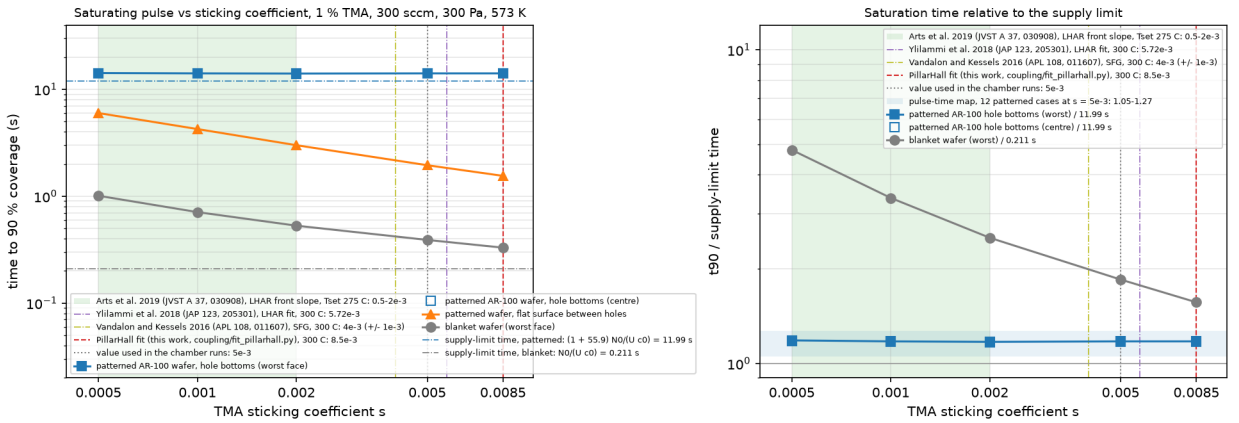


Figure 22: Sensitivity of the saturating TMA pulse to the sticking coefficient at the baseline operating point (1 % TMA, 300 sccm, 300 Pa, 573 K, 15 mm gap). Left: time to 90 % coverage against s for the blanket wafer (worst face), the flat surface between the holes of the AR-100 patterned wafer, and the hole bottoms of the patterned wafer (centre and worst face); dash-dotted horizontal lines are the supply-limit times $(1 + 55.9)\, N_0/(Uc_0) = 11.99\,\mathrm{s}$ and $N_0/(Uc_0) = 0.211\,\mathrm{s}$. Right: the same times divided by their supply-limit time; the blue band is the 1.05–1.27 range of the twelve patterned cases of the pulse-time map at $s = 5 \times 10^{-3}$. In both panels the green band is the Arts et al. [5] range (0.5–2×10^{-3}, set-point 275 °C), and the vertical lines are the Ylilammi et al. [4] fit (5.72×10^{-3}, 300 °C), the Vandalon and Kessels [22] sum-frequency-generation value (4×10^{-3}, 300 °C), the PillarHall fit of this work (8.5×10^{-3}, 300 °C) and the value used in the chamber runs (5×10^{-3}).

not depend on the chamber outboard of the wafer at this Péclet number.

- Robustness checks continued: 200, 300 and 400 °C at fixed sccm and feed fraction give the same pulse times within one write interval (the supply Uc_0 is temperature-independent; temperature acts through the chemistry, held fixed here); 50, 100 and 200 nm holes at fixed open area and aspect ratio saturate within 1 % of each other, so the wall-area ratio is the rule's only geometric input.

- Limitations: isothermal gas (temperature enters only through properties); frozen carrier flow; irreversible Langmuir in the main runs; a first-order desorption term was added for water and shows the purge-tail mechanism (desorbing water holds the exhaust above 1 % for seconds at $k_{\mathrm{des}} = 0.1$ to $1\,\mathrm{s}^{-1}$) and the equilibrium-coverage limit (above $\sim 0.3\,\mathrm{s}^{-1}$ at 1 % H_2O the surface never reaches 90 %), Fig. 9, but k_{des} is a parameter, not a measurement, and a two-site chemisorbed/physisorbed model is the next step; single pattern density (14 % open area, half the wafer patterned); Knudsen $D_K = d\bar{v}/3$ without Clausing correction; s and N_0 are literature-range inputs (the sensitivity to s is quantified above; the patterned-wafer times carry only the site-density uncertainty).

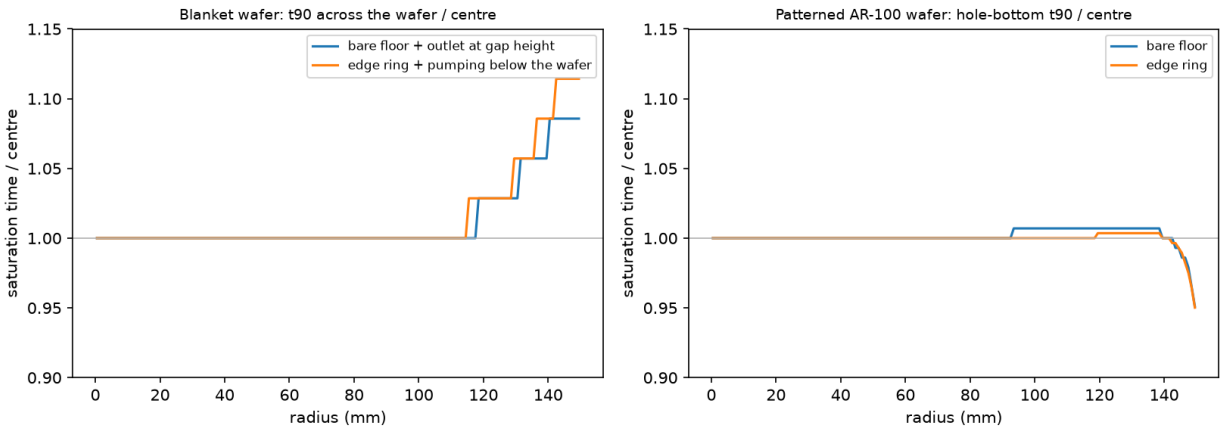


Figure 23: Robustness to the chamber geometry outboard of the wafer, baseline conditions, 1 % TMA. Saturation time across the wafer divided by its value at the centre for the bare floor with the outlet at gap height and for a 25 mm non-absorbing edge ring with the pumping slot 25 mm below the wafer plane. Left: blanket wafer. Right: hole-bottom saturation on the patterned AR-100 wafer. The two geometries differ by at most one write interval.

5 Conclusions

1. On a patterned wafer, the ALD pulse that saturates the deepest feature is set by precursor supply:

$$t_{\mathrm{pulse}} = 1.2\,(1 + A_{\mathrm{wall}}/A_{\mathrm{wafer}})\,\frac{N_0}{U c_0}, \tag{7}$$

   within 25 % over aspect ratios 20–100 and a twenty-fold range of feed fraction, and independent of hole diameter at fixed wall area, of temperature at fixed sccm and fraction, of the chamber geometry outboard of the wafer, of the TMA sticking coefficient across its published range (5×10^{-4} to 8.5×10^{-3}: 1.17 to 1.18 supply-limit times, three write intervals of movement) and, in a full 3D chamber with a one-sided pump, of the pumping arrangement (azimuthal spread of the saturation time under 1 %, within the rule's factor 1.2).

2. The same normalisation puts a measured fluidized-bed ALD run at 1.12 supply-limit times and orders the measured utilization of a rotary-drum reactor with loading; wafers and powders sit on one curve of utilization against surface area spanning five orders of magnitude.

3. The chamber model reproduces the published Fiji F200 case where that case can be represented: the TMA exposure delivered to the wafer per pulse agrees with the authors' three-dimensional CFD to 12 to 22 % and the saturation state of the 25 and 60 ms pulses with the measured growth per cycle; the 2.4 % measured non-uniformity, which comes from an off-axis pulse recirculation, a cold loading door and softer-than-Langmuir kinetics, is not reproduced by an axisymmetric frozen-flow model. The blanket saturation exposure, 0.146 Pa s at the surface, sits inside the 0.05 to 1.3 Pa s spread of kinetic saturation exposures from seven independent sources.

4. A recipe saturated on a blanket wafer under-doses a product wafer by the wall-area ratio, 11 to 56 here. Blanket wafers are the kinetics-limited exception to the rule, not the norm.

5. On a non-uniform layout the pulse is set by the local wall-area ratio of the most demanding region, not by the wafer average: the average under-predicts by 2.3× (dense centre) and 1.2× (aspect-ratio ramp), the local ratio is conservative by 15 to 21 %, and a pattern boundary puts a 26 % saturation gradient inside the array.

6. Real feature shapes and other chemistries stay inside the rule: tapered channel holes cost 1 to 4 % on the supply-limited wafer (the anchored 90 % taper is within 5 % of the straight hole either way); word-line recesses fed through a slit saturate within 4 % of a plain slit with the same wall area up to recess aspect ratio 35, so their cost is in the wall area alone (150 to 189 wafer-areas, 38 to 48 s at 1 %); and the rule is chemistry-agnostic through s, N_0 and c_0, with the HfO_2/TDMAH chamber run at 1.17 times its rule and TiN and W predicted at 4.4 and 57 s per cycle on the same wafer.

7. A 3D voxel Monte Carlo feature solver agrees with converged Machball within 2 % and places the 1D hole model 3 to 11 % fast between AR 100 and AR 20; the 1D slit model with the hydraulic-diameter diffusivity is 24 % slow, and with the parallel-plate correction the word-line recess numbers come within 6 % of the ballistic solver. A second species cannot block the precursor in the gas at Knudsen numbers of 100 to 1,000: for TMA/H_2O the methane byproduct leaves every result as computed, while for chlorides the HCl desorption rate decides between no effect ($k_B \geq 100\,\mathrm{s}^{-1}$), a 10 to 50 % longer pulse ($k_B \sim 1\,\mathrm{s}^{-1}$) and failure of the rule ($k_B \leq 0.1\,\mathrm{s}^{-1}$ or etching).

8. Second-order knobs: pressure up and carrier flow down raise precursor utilization from about 55 % to 95 % at fixed precursor flow; gap and pressure set within-wafer edge lag (5 to 18 %) but not time.

9. The water half-cycle needs the same pulse as TMA on patterned wafers despite a 50× lower sticking coefficient, and coats deep features more evenly; reversible adsorption, not chamber flushing, governs its purge.

10. In fab units the rule's pulse gives 99 % step coverage as well as 90 %, because the saturation front is 0.25 to 0.3 s wide at every aspect ratio; 50 cycles of the blanket recipe leave 1.4 nm of a 5.15 nm film on top of an AR-100 array and nothing at the bottom. The Al_2O_3 cycle on the AR-100 wafer is 3.1 times the blanket cycle (37.6 against 12.2 s; 1.9 against 5.5 wafers per hour for 5 nm), its pulses are three quarters of the cycle, and above a wall-area ratio of about 10 throughput falls as $1/(1 + A_{\mathrm{wall}}/A_{\mathrm{wafer}})$; the purge is set by water desorption, not by the chamber, which clears to 1 % in 2 s while a 10 μm hole empties in microseconds.

11. Plasma ALD adds a hard ceiling: above a wall recombination probability of about 3×10^{-3}, aspect ratio 100 is unreachable by dose. Separated-chemistry (spatial) architectures beat temporal pulsing by the same wall-area factor the rule predicts.

12. All of this runs on a laptop in open-source tools, with the feature model verified against two independent ballistic descriptions and fitted to public measurements, and every number traceable to a script in the repository.

Acknowledgements

Validation data by R. L. Puurunen and O. M. E. Ylivaara (Aalto University, Zenodo 3974438, CC BY 4.0). Machball by A. Yanguas-Gil and J. W. Elam (Argonne). OpenFOAM v2606 (OpenCFD) via the gerlero/openfoam-app build.

Code and data availability

OpenFOAM v2606 cases, the `aldWallKinetics` boundary-condition library, the Python `aldconf` solver and its command line (`python -m aldconf` evaluates the rule, the surrogate and the Pareto front), tests and all figures: `https://github.com/Soham-ChemE/ald-conformality`. The repository also ships a layout-to-recipe tool on its live page (pattern zones in, per-zone local rule and recipe pulse out, calibrated on the two non-uniform runs of Section 3.9) and continuous integration that rebuilds the boundary conditions and reruns a regression case in an OpenFOAM container (118 tests). The kinetics behind the OpenFOAM boundary conditions live in `lib/aldkin`, a C-ABI library with Fluent and plain-C hooks, and the voxel Monte Carlo feature solver of Section 2.6 in `lib/aldmc`. Validation data: Zenodo 3974438 (CC BY 4.0).

References

[1] R. G. Gordon, D. Hausmann, E. Kim and J. Shepard, "A kinetic model for step coverage by atomic layer deposition in narrow holes or trenches," *Chem. Vap. Deposition* **9**(2), 73–78 (2003). doi:10.1002/cvde.200390005.

[2] A. Yanguas-Gil and J. W. Elam, "Self-limited reaction-diffusion in nanostructured substrates: surface coverage dynamics and analytic approximations to ALD saturation times," *Chem. Vap. Deposition* **18**(1–3), 46–52 (2012). doi:10.1002/cvde.201106938.

[3] A. Yanguas-Gil and J. W. Elam, *Machball: reactive transport inside nanostructured and high-surface-area materials with self-limited surface kinetics*, documentation, Argonne National Laboratory, `https://machball.readthedocs.io/` (accessed November 2026). The ballistic transport model is described in A. Yanguas-Gil and J. W. Elam, "A Markov chain approach to simulate atomic layer deposition chemistry and transport inside nanostructured substrates," *Theor. Chem. Acc.* **133**, 1465 (2014), doi:10.1007/s00214-014-1465-x.

[4] M. Ylilammi, O. M. E. Ylivaara and R. L. Puurunen, "Modeling growth kinetics of thin films made by atomic layer deposition in lateral high-aspect-ratio structures," *J. Appl. Phys.* **123**, 205301 (2018). doi:10.1063/1.5028178.

[5] K. Arts, V. Vandalon, R. L. Puurunen, M. Utriainen, F. Gao, W. M. M. Kessels and H. C. M. Knoops, "Sticking probabilities of H_2O and $Al(CH_3)_3$ during atomic layer deposition of Al_2O_3 extracted from their impact on film conformality," *J. Vac. Sci. Technol. A* **37**, 030908 (2019). doi:10.1116/1.5093620.

[6] K. Arts, M. Utriainen, R. L. Puurunen, W. M. M. Kessels and H. C. M. Knoops, "Film conformality and extracted recombination probabilities of O atoms during plasma-assisted atomic layer deposition of SiO_2, TiO_2, Al_2O_3, and HfO_2," *J. Phys. Chem. C* **123**(44), 27030–27035 (2019). doi:10.1021/acs.jpcc.9b08176.

[7] K. Arts, S. Deijkers, R. L. Puurunen, W. M. M. Kessels and H. C. M. Knoops, "Oxygen recombination probability data for plasma-assisted atomic layer deposition of SiO_2 and TiO_2," *J. Phys. Chem. C* **125**(15), 8244–8252 (2021). doi:10.1021/acs.jpcc.1c01505.

[8] V. Cremers, R. L. Puurunen and J. Dendooven, "Conformality in atomic layer deposition: current status overview of analysis and modelling," *Appl. Phys. Rev.* **6**, 021302 (2019). doi:10.1063/1.5060967.

[9] J. Yim, O. M. E. Ylivaara, M. Ylilammi, V. Korpelainen, E. Haimi, E. Verkama, M. Utriainen and R. L. Puurunen, "Saturation profile based conformality analysis for atomic layer deposition: aluminum oxide in lateral high-aspect-ratio channels," *Phys. Chem. Chem. Phys.* **22**(40), 23107–23120 (2020). doi:10.1039/D0CP03358H.

[10] J. Marles, O. M. E. Ylivaara and R. L. Puurunen, "ALD Al_2O_3 thickness profile in microscopic rectangular channel: TMA/H_2O, 300 °C, 500 cycles, PillarHall LHAR3 (500 nm gap)," dataset, Zenodo, version 1.0.0 (2021), CC BY 4.0. doi:10.5281/zenodo.3974438.

[11] J. Järvilehto, J. A. Velasco, J. Yim, C. Gonsalves and R. L. Puurunen, "Simulation of conformality of ALD growth inside lateral channels: comparison between a diffusion–reaction model and a ballistic transport–reaction model," *Phys. Chem. Chem. Phys.* **25**(34), 22952–22964 (2023). doi:10.1039/D3CP01829F.

[12] Z. Lu, A. Yanguas-Gil, D. Kang, P. Darapaneni, A. U. Mane, C. L. Marshall and J. W. Elam, "Scalable synthesis of supported catalysts using fluidized bed atomic layer deposition," *J. Vac. Sci. Technol. A* **40**(4), 042404 (2022). doi:10.1116/6.0001891; OSTI 1902316.

[13] A. Yanguas-Gil and J. W. Elam, "Modeling scale-up of particle coating by atomic layer deposition," arXiv:2408.13116 (2024). doi:10.48550/arXiv.2408.13116.

[14] M. W. Coile, M. J. Young, J. A. Libera, A. U. Mane and J. W. Elam, "High-capacity rotary drum for atomic layer deposition onto powders and small mechanical parts in a hot-walled viscous flow reactor," *J. Vac. Sci. Technol. A* **38**(5), 052403 (2020). doi:10.1116/6.0000274; OSTI 1660751.

[15] U. Kelkar, "Semiconductor equipment and processes need digital twins," Applied Materials blog, 5 July 2023. https://www.appliedmaterials.com/us/en/blog/blog-posts/semiconductor-equipment-and-processes-need-digital-twins.html (accessed November 2026).

[16] Applied Materials, Inc., "Applied Materials introduces high-performance ALD technology for the 3D era," press release (Applied Olympia ALD system), Santa Clara, CA, 13 July 2015. https://ir.appliedmaterials.com/news-releases/news-release-details/applied-materials-introduces-high-performance-ald-technology-3d (accessed November 2026).

[17] Applied Materials, Inc., "Applied Materials unveils deposition and selective etch systems to advance 3D chip scaling," press release (Centris Spectral SiN ALD system), 15 June 2026. https://www.appliedmaterials.com/us/en/newsroom/press-releases/061526-applied-materials-unveils-deposition-selective-etch-systems-to-advance-3d-chip-scaling.html (accessed November 2026).

[18] N. Li, V. Nguyen, M. Balseanu, L.-Q. Xia, S. D. Marcus, H. Yang and K. Tanaka (Applied Materials, Inc.), "Seamless gap-fill with spatial atomic layer deposition," US patent application publication US 2015/0255324 A1, published 10 September 2015.

[19] OpenCFD Ltd., *OpenFOAM v2606*, June 2026, https://www.openfoam.com; built for macOS with the gerlero/openfoam-app distribution, https://github.com/gerlero/openfoam-app (accessed November 2026).

[20] G. P. Gakis, H. Vergnes, E. Scheid, C. Vahlas, B. Caussat and A. G. Boudouvis, "Computational fluid dynamics simulation of the ALD of alumina from TMA and H_2O in a commercial reactor," *Chem. Eng. Res. Des.* **132**, 795–811 (2018). doi:10.1016/j.cherd.2018.02.031.

[21] G. P. Gakis, H. Vergnes, E. Scheid, C. Vahlas, A. G. Boudouvis and B. Caussat, "Detailed investigation of the surface mechanisms and their interplay with transport phenomena in alumina atomic layer deposition from TMA and water," *Chem. Eng. Sci.* **195**, 399–412 (2019). doi:10.1016/j.ces.2018.09.037.

[22] V. Vandalon and W. M. M. Kessels, "What is limiting low-temperature atomic layer deposition of Al_2O_3? A vibrational sum-frequency generation study," *Appl. Phys. Lett.* **108**, 011607 (2016). doi:10.1063/1.4939654.

[23] V. Vandalon and W. M. M. Kessels, "Revisiting the growth mechanism of atomic layer deposition of Al_2O_3: a vibrational sum-frequency generation study," *J. Vac. Sci. Technol. A* **35**, 05C313 (2017). doi:10.1116/1.4993597.

[24] V. Vandalon and W. M. M. Kessels, "Initial growth study of atomic-layer deposition of Al_2O_3 by vibrational sum-frequency generation," *Langmuir* **35**(32), 10374–10382 (2019). doi:10.1021/acs.langmuir.9b01600.

[25] R. L. Puurunen, "Surface chemistry of atomic layer deposition: a case study for the trimethylaluminum/water process," *J. Appl. Phys.* **97**, 121301 (2005). doi:10.1063/1.1940727.

[26] C. Soto and W. T. Tysoe, "The reaction pathway for the growth of alumina on high surface area alumina and in ultrahigh vacuum by a reaction between trimethyl aluminum and water," *J. Vac. Sci. Technol. A* **9**, 2686–2695 (1991). doi:10.1116/1.577226.

[27] A. C. Dillon, A. W. Ott, J. D. Way and S. M. George, "Surface chemistry of Al_2O_3 deposition using $Al(CH_3)_3$ and H_2O in a binary reaction sequence," *Surf. Sci.* **322**, 230–242 (1995). doi:10.1016/0039-6028(95)90033-0.

[28] T. Reiter, L. F. Aguinsky, F. Rodrigues, J. Weinbub, A. Hössinger and L. Filipovic, "Modeling the impact of incomplete conformality during atomic layer processing," *Solid-State Electron.* **211**, 108816 (2024). doi:10.1016/j.sse.2023.108816.

[29] D. Blaschke, F. Munnik, J. Grenzer, L. Rebohle, H. Schmidt, P. Zahn and S. Gemming, "A correlation study of layer growth rate, thickness uniformity, stoichiometry, and hydrogen impurity level in HfO_2 thin films grown by ALD between 100 °C and 350 °C," *Appl. Surf. Sci.* **506**, 144188 (2020). doi:10.1016/j.apsusc.2019.144188.

[30] S. Gieraltowska, L. Wachnicki, P. Dluzewski, B. S. Witkowski, M. Godlewski and E. Guziewicz, "Atomic layer deposition of HfO_2 films using TDMAH and water or ammonia water," *Materials* **16**(11), 4077 (2023). doi:10.3390/ma16114077.

[31] D. M. Hausmann, E. Kim, J. Becker and R. G. Gordon, "Atomic layer deposition of hafnium and zirconium oxides using metal amide precursors," *Chem. Mater.* **14**(10), 4350–4358 (2002). doi:10.1021/cm020357x.

[32] A. Delabie, R. L. Puurunen, B. Brijs, M. Caymax, T. Conard, B. Onsia, O. Richard, W. Vandervorst, C. Zhao, M. M. Heyns, M. Meuris, M. M. Viitanen, H. H. Brongersma, M. de Ridder, L. V. Goncharova, E. Garfunkel, T. Gustafsson and W. Tsai, "Atomic layer deposition of hafnium oxide on germanium substrates," *J. Appl. Phys.* **97**, 064104 (2005). doi:10.1063/1.1856221.

[33] S. A. Kinnunen, J. Malm, K. Arstila, M. Lahtinen and T. Sajavaara, "Characterization of ALD grown Ti_xAl_yN and Ti_xAl_yC thin films," *Nucl. Instrum. Methods Phys. Res. B* **406**, 152–155 (2017). doi:10.1016/j.nimb.2016.12.032.

[34] A. Satta, J. Schuhmacher, C. M. Whelan, W. Vandervorst, S. H. Brongersma, G. P. Beyer, K. Maex, A. Vantomme, M. M. Viitanen, H. H. Brongersma and W. F. A. Besling, "Growth mechanism and continuity of atomic layer deposited TiN films on thermal SiO_2," *J. Appl. Phys.* **92**(12), 7641–7646 (2002). doi:10.1063/1.1522485.

[35] J. W. Elam, C. E. Nelson, R. K. Grubbs and S. M. George, "Kinetics of the WF_6 and Si_2H_6 surface reactions during tungsten atomic layer deposition," *Surf. Sci.* **479**, 121–135 (2001). doi:10.1016/S0039-6028(01)00969-4.

[36] J. W. Klaus, S. J. Ferro and S. M. George, "Atomic layer deposition of tungsten using sequential surface chemistry with a sacrificial stripping reaction," *Thin Solid Films* **360**, 145–153 (2000). doi:10.1016/S0040-6090(99)01076-7.

[37] D. Lee and C. Shin, "Impact of stacking-up and scaling-down bit cells in 3D NAND on their threshold voltages," *Micromachines* **13**(7), 1139 (2022). doi:10.3390/mi13071139.

[38] Lam Research Corp., "Lam Research introduces Lam Cryo 3.0 cryogenic etch technology to accelerate scaling of 3D NAND for the AI era," press release, Fremont, CA, 31 July 2024. https://www.nasdaq.com/press-release/lam-research-introduces-lam-cryotm-30-cryogenic-etch-technology-accelerate-scaling-3d (accessed November 2026).

[39] EE Times Japan, article on Lam Research's third-generation Cryo etch for 1000-layer NAND (in Japanese), 2 September 2024. https://eetimes.itmedia.co.jp/ee/spv/2409/02/news119.html (accessed November 2026).

[40] N. Keller et al. (Onto Innovation), "Measuring the critical W-recess in 3D NAND," *Semiconductor Digest*, July 2023, p. 31. https://ontoinnovation.com/writable/files/Measuring-the-Critical-W-recess-in-3D-NAND_Semi-Digest_July_2023.pdf (accessed November 2026).

[41] 3DInCites, analysis of TechInsights' cross-sections of Samsung's 86 Gbit 32-layer V-NAND, December 2014. https://www.3dincites.com/?p=8892 (accessed November 2026).

[42] M. Ritala, M. Leskelä, E. Rauhala and P. Haussalo, "Atomic layer epitaxy growth of TiN thin films," *J. Electrochem. Soc.* **142**(8), 2731–2737 (1995). doi:10.1149/1.2050083.

[43] H. Tiznado and F. Zaera, "Surface chemistry in the atomic layer deposition of TiN films from $TiCl_4$ and ammonia," *J. Phys. Chem. B* **110**(27), 13491–13498 (2006). doi:10.1021/jp062019f.

[44] M. Q. Snyder, B. A. McCool, J. DiCarlo, C. P. Tripp and W. J. DeSisto, "An infrared study of the surface chemistry of titanium nitride atomic layer deposition on silica from $TiCl_4$ and NH_3," *Thin Solid Films* **514**, 97–102 (2006). doi:10.1016/j.tsf.2006.03.013.

[45] J. Leem, I. Park, Y. Li, W. Zhou, Z. Jin, S. Shin and Y.-S. Min, "Role of HCl in atomic layer deposition of TiO_2 thin films from titanium tetrachloride and water," *Bull. Korean Chem. Soc.* **35**(4), 1195–1201 (2014). doi:10.5012/bkcs.2014.35.4.1195.

[46] S. Kim, H. An, S. Oh, J. Jung, B. Kim, S. K. Nam and S. Han, "Atomistic kinetic Monte Carlo simulation on atomic layer deposition of TiN thin film," *Comput. Mater. Sci.* **213**, 111620 (2022). doi:10.1016/j.commatsci.2022.111620.

[47] J. W. Elam, C. E. Nelson, M. A. Tolbert and S. M. George, "Adsorption and desorption of HCl on a single-crystal α-Al_2O_3(0001) surface," *Surf. Sci.* **450**, 64–77 (2000). doi:10.1016/S0039-6028(99)01247-9.

[48] A. Lostier, Y. Segev, T. Kislev, G. Schwartz Roitman, N. Locoge and M. N. Romanias, "Uptake of stratospheric species on minerals proposed for stratospheric aerosol injection," arXiv:2605.14826 (2026). doi:10.48550/arXiv.2605.14826.

[49] A. S. Berman, "Free molecule transmission probabilities," *J. Appl. Phys.* **36**, 3356 (1965). doi:10.1063/1.1702984.

[50] S. M. George, "Atomic layer deposition: an overview," *Chem. Rev.* **110**(1), 111–131 (2010). doi:10.1021/cr900056b.